

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

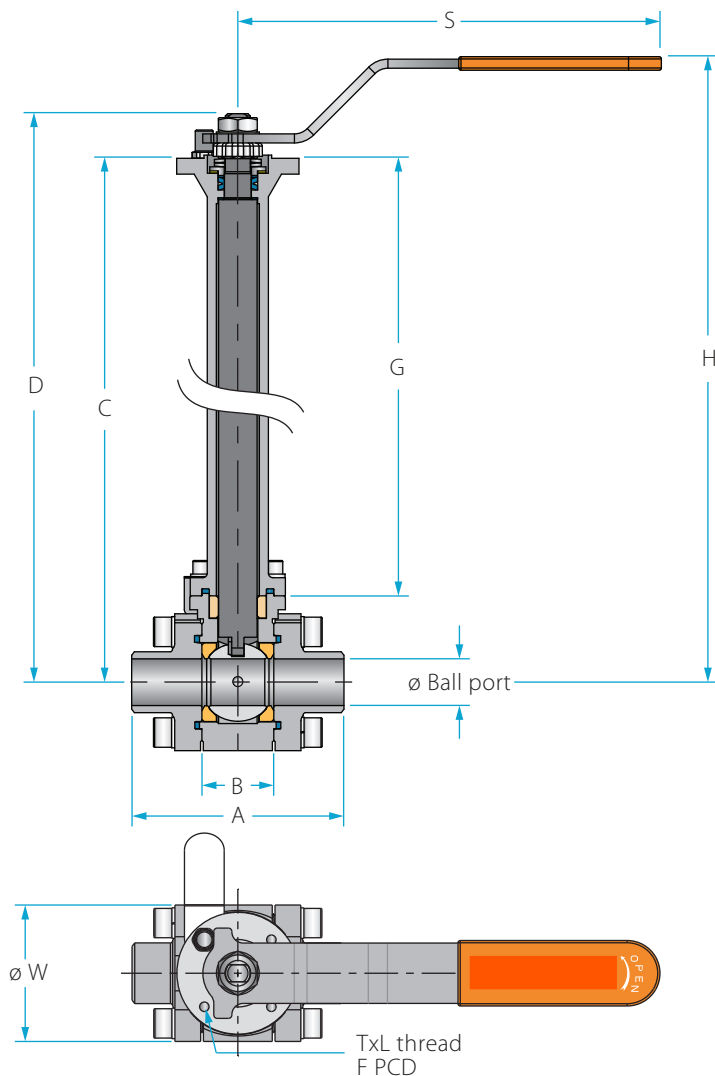
High
pressure

Multiport

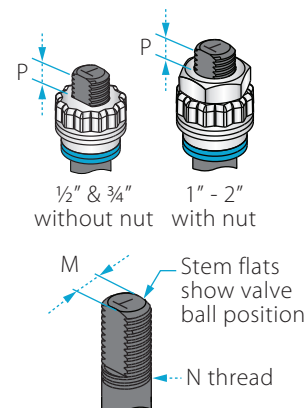
Ordering
code system

Size ¼" - 2" | DN8-DN50 | Class 600 | C47W/C47C⁽¹⁾ Series

Valve dimensions

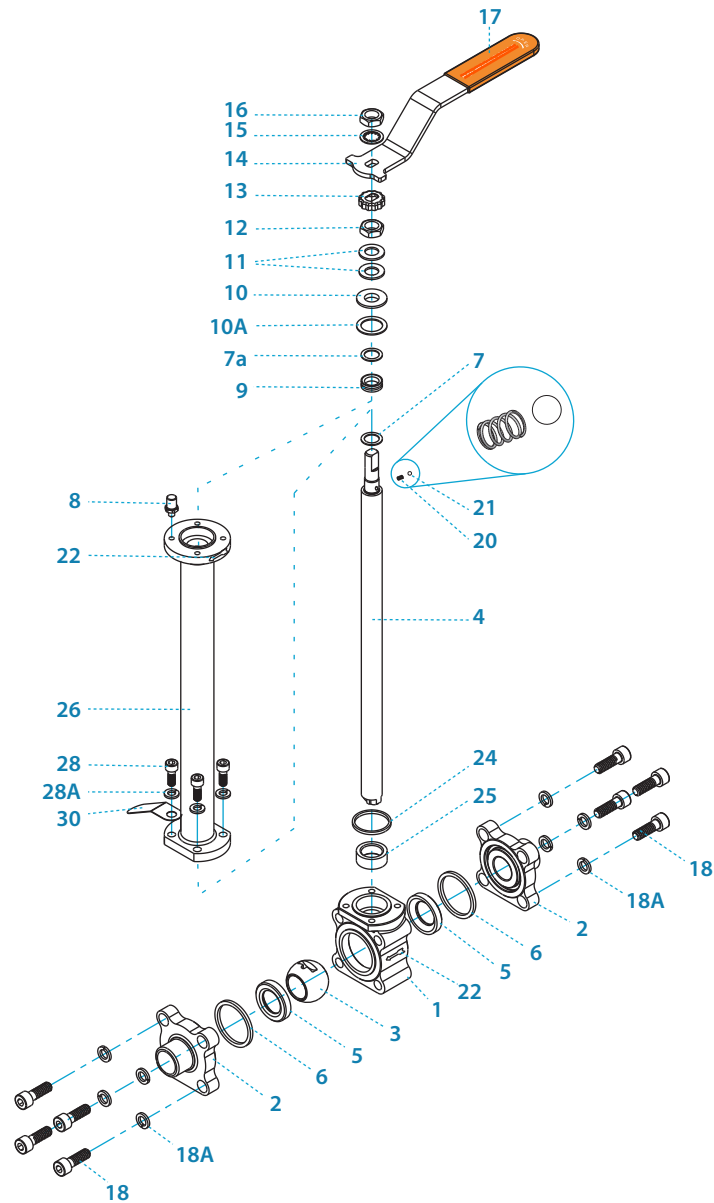


Preparation for actuation



Std. port	Full port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	N	P	F	TxL	Weight kg/lb		Kv
						C	D	G	H	C	D	G	H								Std.	6.0"	Cv
DN15	DN10	mm	11.15	65.80	20.60	333.00	341.90	304.00	366.00	181.40	190.30	152.40	214.40	150.00	46.00	5.50	3⁄8"	7.20 (F03)	36.00	M5X7	1.60	1.28	7
½"	⅜"	inch	0.44	2.59	0.81	13.11	13.46	11.97	14.41	7.14	7.49	6.00	8.44	5.91	1.81	0.22	UNF	0.28	1.42			3.53	2.82
DN20	DN15	mm	14.30	70.55	24.55	335.40	344.30	304.00	368.00	183.80	192.70	152.40	216.40	150.00	52.00	5.50	3⁄8"	7.20 (F03)	36.00	M5X7	1.80	1.48	10
¾"	½"	inch	0.56	2.78	0.97	13.20	13.56	11.97	14.49	7.24	7.59	6.00	8.52	5.91	2.05	0.22	UNF	0.28	1.42			3.97	3.26
DN25	DN20	mm	20.60	93.65	31.75	342.15	359.65	304.00	383.00	190.55	208.05	152.40	231.40	187.00	60.50	7.54	7⁄16"	7.20 (F04)	42.00	M5X7	3.00	2.65	28
1"	¾"	inch	0.81	3.69	1.25	13.47	14.16	11.97	15.08	7.50	8.19	6.00	9.11	7.32	2.38	0.30	UNF	0.28	1.65			6.61	5.84
DN32	DN25	mm	25.40	108.25	41.25	346.65	364.15	304.00	388.00	195.05	212.55	152.40	236.40	187.00	69.00	7.54	7⁄16"	7.20 (F04)	42.00	M5X10	3.70	3.35	37
1¼"	1"	inch	1.00	4.25	1.62	13.65	14.34	11.97	15.28	7.68	8.37	6.00	9.31	7.32	2.72	0.30	UNF	0.28	1.65			8.16	7.39
DN40	DN32	mm	31.80	115.45	48.25	347.55	377.05	304.00	400.00	195.95	225.45	152.40	248.40	237.00	79.20	8.71	9⁄16"	8.00 (F05)	50.00	M6X8	5.20	4.65	70
1½"	1¼"	inch	1.25	4.55	1.90	13.68	14.84	11.97	15.75	7.71	8.88	6.00	9.78	9.29	3.12	0.34	UNF	0.31	1.97			11.46	10.25
DN50	DN40	mm	38.10	127.10	56.30	352.25	381.75	304.00	405.00	200.65	230.15	152.40	253.40	237.00	90.70	8.71	9⁄16"	8.50 (F05)	50.00	M6X8	6.10	5.53	103
2"	1½"	inch	1.50	5.00	2.22	13.87	15.03	11.97	15.94	7.90	9.06	6.00	9.98	9.29	3.57	0.34	UNF	0.33	1.97			13.45	12.19

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A351 CF8M, A494 M35-1, Bronze RG5	2
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	2
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1
13	Locking clip	A167 304	1

Item	Description	Material specifications	Qty.
14	Handle	A240 430	1
15	Serrated washer	A240 410	1
16	Handle nut	EN3506-2 A4-80, A194 Gr 8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	8
18A	Spring washer	DIN 127 A2	8
20	Anti-static spring	A313 302	1
21	Anti-static plunger	A479 304	1
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	A193 B8, DIN 912 A2-70	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

⁽¹⁾ C47C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C47C design.

* Repair kit components

** Only with HermetiX™ fire safe design

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

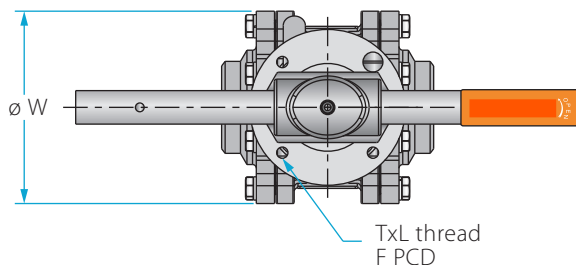
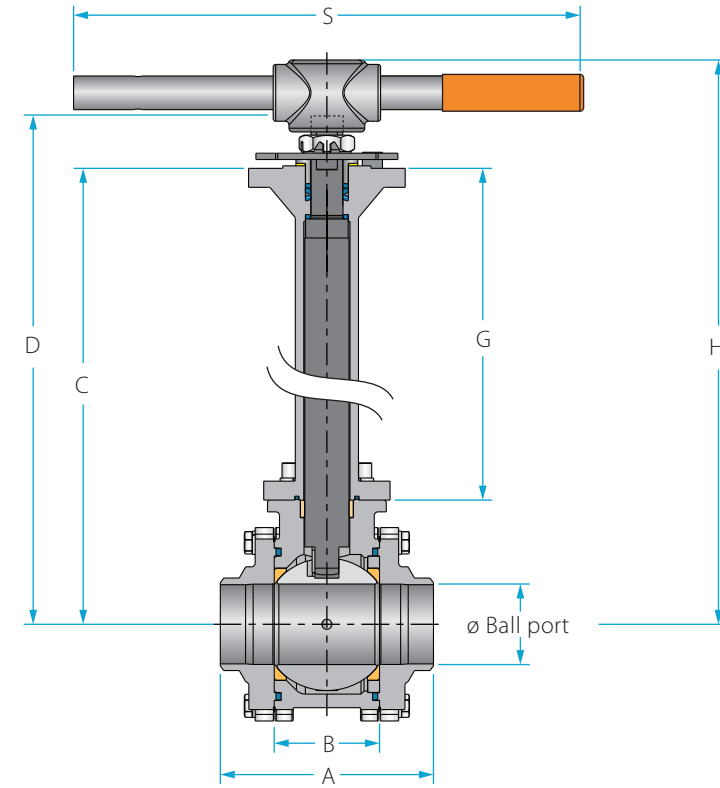
High
pressure

Multiport

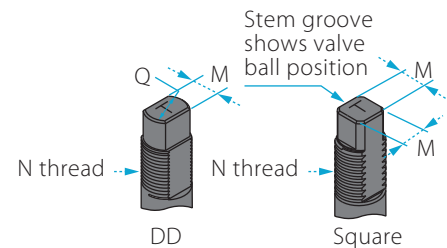
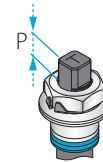
Ordering
code system

Size 2½" - 6" | DN65-DN150 | Class 300 | C47W/C47C⁽¹⁾ Series

Valve dimensions



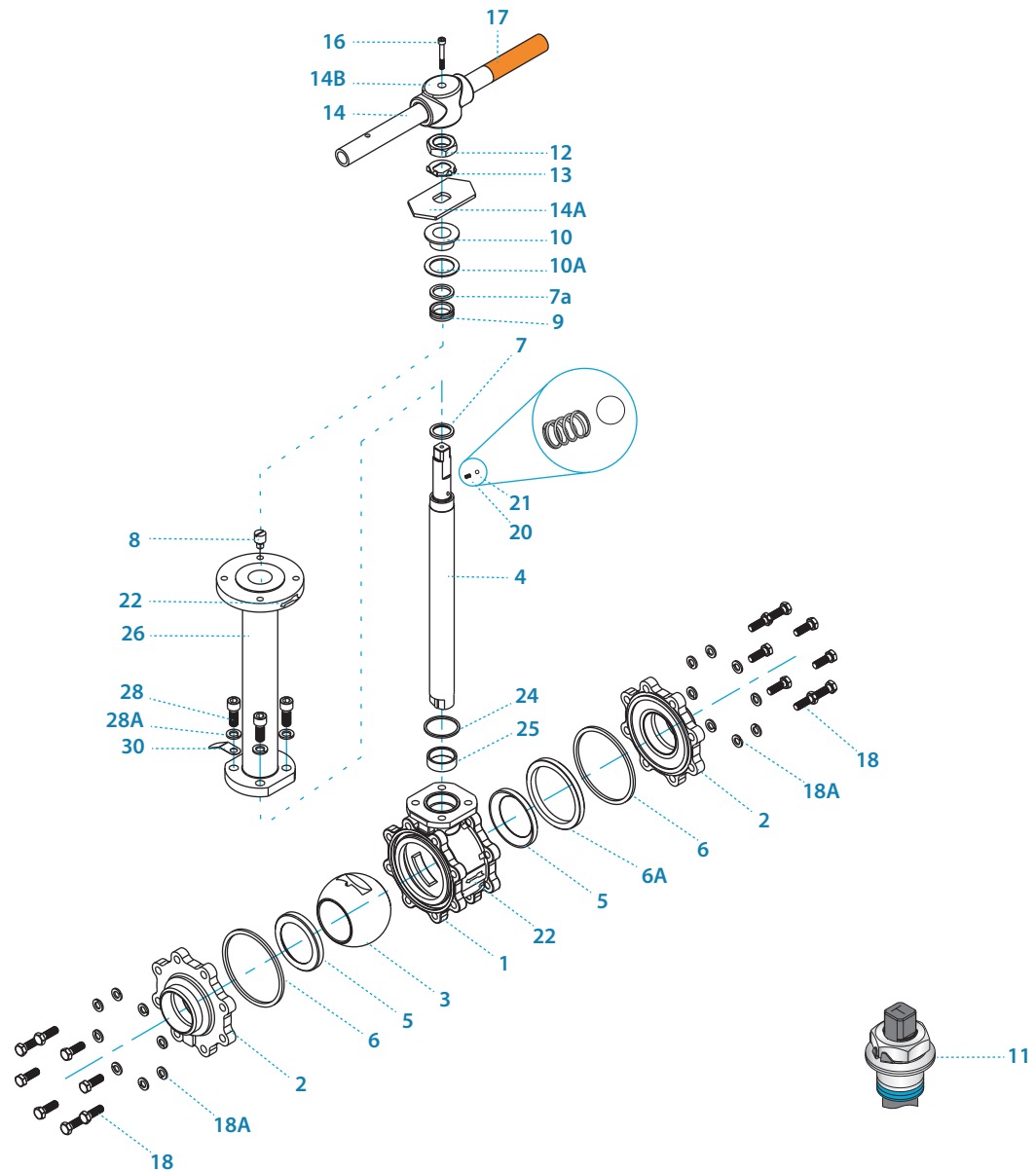
Preparation for actuation



Std. port	Full port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	MDD	N	P	Q	F	TxL	Weight kg/lb		Kv	
						C	D	G	H	C	D	G	H										Std.	6.0"		Cv
DN65	DN50	mm	50.80	157.95	72.55	397.50	439.10	327.50	464.00	222.40	264.00	152.40	288.90	401.00	108.00	13.90	13.90	M20x2.5	13.15	20.00	(F07)	70.00	M8x8	12.00	10.60	205
2½"	2"	inch	2.00	6.22	2.86	15.65	17.29	12.89	18.27	8.76	10.39	6.00	11.37	15.79	4.25	0.55	0.55		0.52	0.79		2.76		26.46	23.37	240
DN80	DN65	mm	63.50	169.35	83.25	441.80	488.40	343.50	528.00	250.70	297.30	152.40	336.90	401.00	153.00	18.90	15.90	1"-14	16.70	22.70	(F10)	102.00	M10x15	20.00	18.80	385
3"	2½"	inch	2.50	6.67	3.28	17.39	19.23	13.52	20.79	9.87	11.70	6.00	13.26	15.79	6.02	0.74	0.63	UNS-2A	0.66	0.89		4.02		44.09	41.45	450
DN100	DN80	mm	82.60	213.60	108.80	457.60	504.20	343.50	544.00	266.50	313.10	152.40	352.90	610.00	191.50	18.90	15.90	1"-14	16.70	22.70	(F10)	102.00	M10x15	30.50	29.30	615
4"	3"	inch	3.25	8.41	4.28	18.02	19.85	13.52	21.42	10.49	12.33	6.00	13.89	24.02	7.54	0.74	0.63	UNS-2A	0.66	0.89		4.02		67.24	64.60	720
	DN100	mm	100.00	239.00	123.00	466.80	513.40	343.50	555.00	275.70	322.30	152.40	363.90	610.00	217.00	18.90	15.90	1"-14	16.70	22.70	(F10)	102.00	M10x15	37.30	36.10	744
	4"	inch	3.94	9.41	4.84	18.38	20.21	13.52	21.85	10.85	12.69	6.00	14.33	24.02	8.54	0.74	0.63	UNS-2A	0.66	0.89		4.02		82.23	79.59	870
DN150		mm	111.10	346.10	146.10	542.40	611.90	385.00	665.00	309.80	379.30	152.40	432.40	916.00	266.00	28.45	23.75	1½"-12	26.20	35.20	(F12)	125.00	M12x15	75.00	71.00	872
6"		inch	4.37	13.63	5.75	21.35	24.09	15.16	26.18	12.20	14.93	6.00	17.02	36.06	10.47	1.12	0.94	UNS-1A	1.03	1.39		4.92		165.35	156.53	1020

⁽¹⁾ C47C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A351 CF8M, A494 M35-1, Bronze RG5	2
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	2
6A	Support ring	A351 CF8M	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1
13	Tab lock washer	A240 304	1

⁽¹⁾ C47C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C47C design.

Item	Description	Material specifications	Qty.
14	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	8
18A	Spring washer	DIN 127 A2	8
20	Anti-static spring	A313 302	2
21	Anti-static plunger	A479 304	2
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	A193 B8, DIN 912 A2-70	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

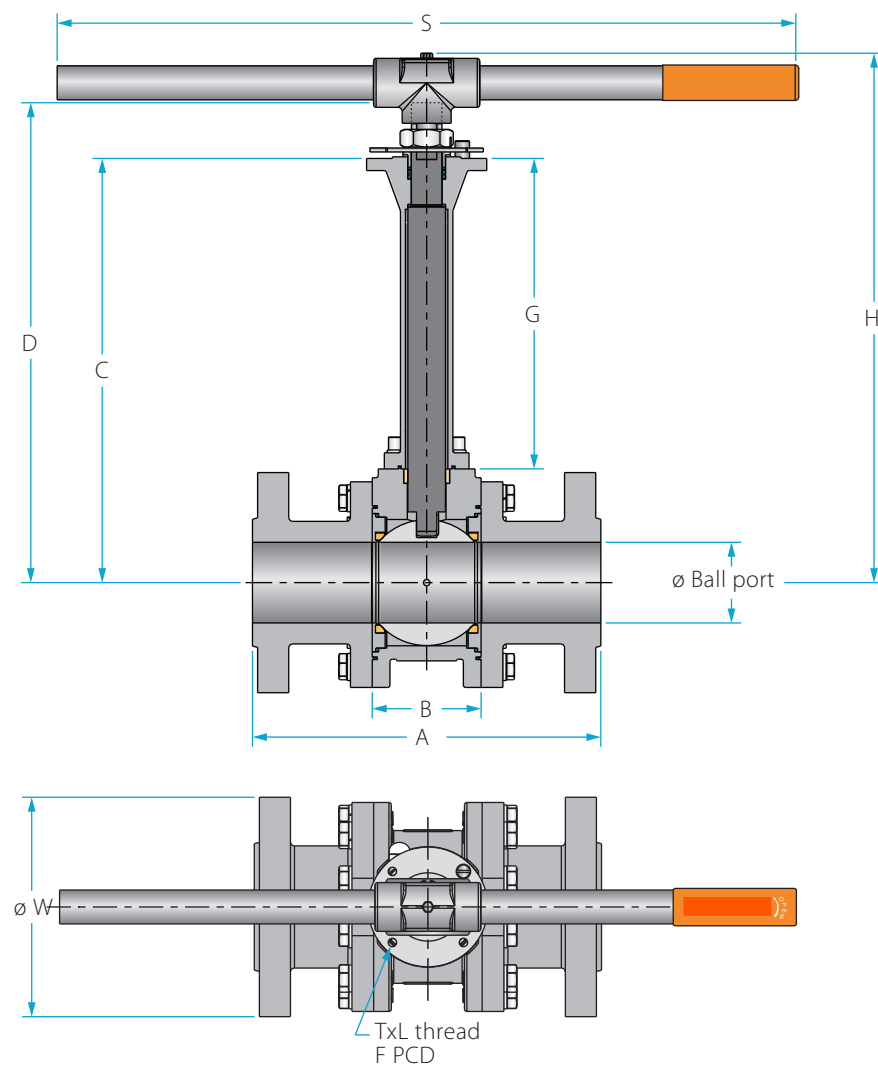
High
pressure

Multiport

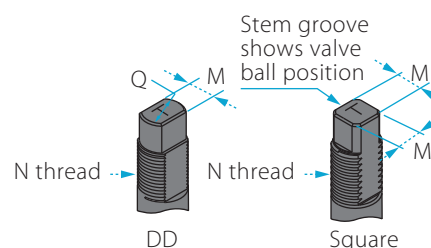
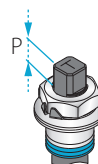
Ordering
code system

Size 2"- 8" | DN50-DN200 | Class 600 | C26W/C26C⁽¹⁾ Series

Valve dimensions



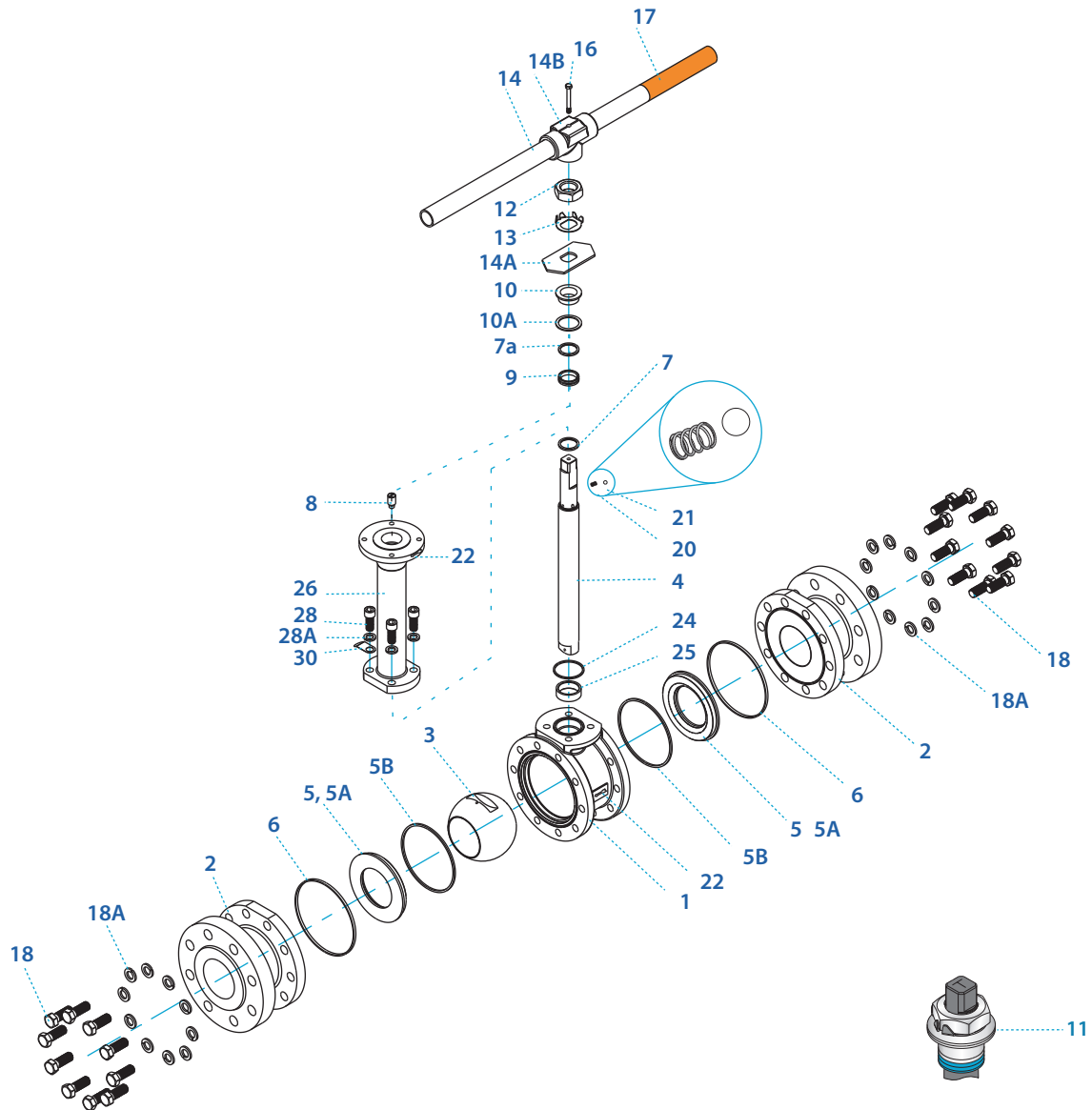
Preparation for actuation



Full Port	Unit	Ball port	A		B	Standard design				6.0" bonnet				S	W	M	MDD	N	P	Q	F	TxL	Weight Kg/lb (Welded)		Kv	
			Welded	#600		C	D	G	H	C	D	G	H										Std.	6.0"		
																										Cv
DN50	mm	51.05	200.00	292.00	75.00	432.50	479.10	345.00	431.00	239.90	286.50	152.40	238.40	401.00	165.00	18.90	15.90	1" - 14	16.70	22.70	(F10)	102.00	M10X15	25.00	22.00	42
2"	inch	2.01	7.87	11.50	2.95	17.03	18.86	13.58	16.97	9.44	11.28	6.00	9.39	15.79	6.50	0.74	0.63	UNS-2A	0.66	0.89		4.02		55.00	48.00	49
DN80	mm	80.00	356.00	356.00	110.00	511.50	580.60	385.00	632.00	278.90	348.00	152.40	399.40	916.00	210.00	28.45	23.75	1-½"	26.20	35.20	(F12)	125.00	M12X15	50.00	45.00	98
3"	inch	3.15	14.02	14.02	4.33	20.14	22.86	15.16	24.88	10.98	13.70	6.00	15.72	36.06	8.27	1.12	0.94	UNF-2A	1.03	1.39		4.92		110.00	99.00	115
DN100	mm	100.00	400.00	432.00	135.00	527.30	596.40	385.00	646.50	294.70	363.80	152.40	413.90	916.00	275.00	28.45	23.75	1-½"	26.20	35.20	(F12)	125.00	M12X15	75.00	70.00	180
4"	inch	3.94	15.75	17.01	5.31	20.76	23.48	15.16	25.45	11.60	14.32	6.00	16.30	36.06	10.83	1.12	0.94	UNF-2A	1.03	1.39		4.92		165.00	154.00	211
DN150	mm	150.00	490.00	559.00	190.00	579.00	691.65	385.00	-	346.40	459.05	152.40	-	-	355.00	35.92	35.92	2"	40.00	46.50	(F14)	140.00	M16X20	155.00	140.00	431
6"	inch	5.91	19.29	22.01	7.48	22.80	27.23	15.16	-	13.64	18.07	6.00	-	-	13.98	1.41	1.41	UN-2A	1.57	1.83		5.51		341.00	308.00	504
DN200	mm	200.00	500.00	660.00	275.00	630.00	742.65	385.00	-	397.40	510.05	152.40	-	-	420.00	45.90	45.90	2-¾"	50.00	59.00	(F16)	165.00	M20X30	400.00	380.00	797
8"	inch	7.87	19.69	25.98	10.83	24.80	29.24	15.16	-	15.65	20.08	6.00	-	-	16.54	1.81	1.81	UN-2A	1.97	2.32		6.50		880.00	836.00	932

⁽¹⁾ Due to high valve torque, pipe handle cannot be used. A manual gear or automation means should be used to operate the valve.

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A479 316/316L, A351 CF8M, A494 M35-1, Bronze RG5	2
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
5A*	Seat housing	A351 CF8M, A494 M35-1, Bronze RG5	2
5B*	Seat seal	PTFE, Graphite	2
6*	Body seal	PTFE, Graphite	2
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1

⁽¹⁾ C26C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C26C design.

Item	Description	Material specifications	Qty.
13	Tab lock washer	A240 304	1
14***	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	20-32
18A	Spring washer	DIN 127 A2	20-32
20	Anti-static spring	A313 302	2
21	Anti-static plunger	A479 304	2
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, CF8M	1
28	Bolt	A193 B8, DIN 912 A2-70	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

*** Gear operator should be used for size 6" DN150 and above (handle components are not included)

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

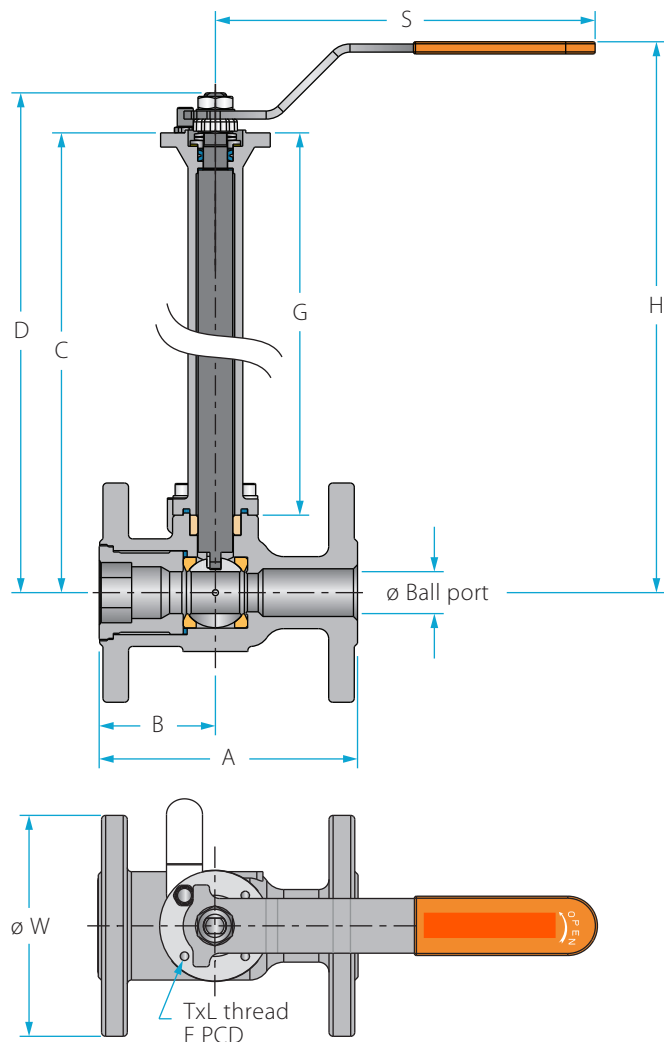
High
pressure

Multiport

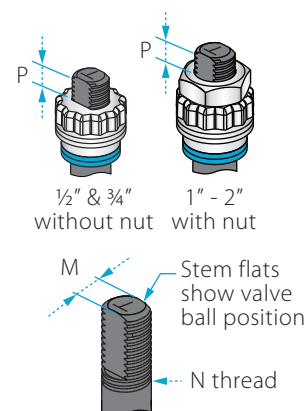
Ordering
code system

Size 1/2" - 2" | DN15-DN50 | ANSI Class 150 | C31W/C31C⁽¹⁾ Series

Valve dimensions

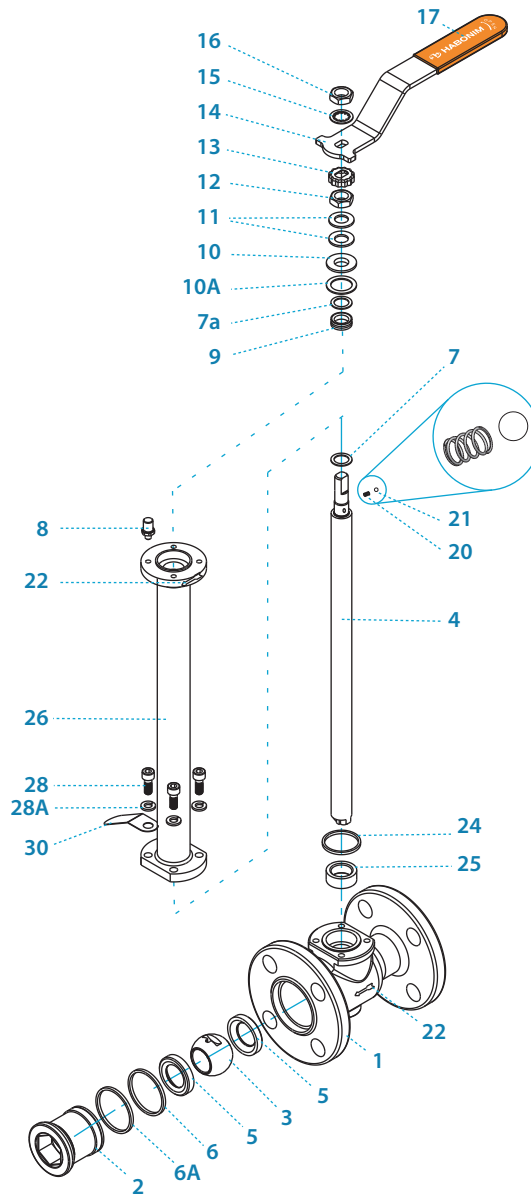


Preparation for actuation



Std. port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	N	P	F	TxL	Weight kg/lb		Kv
					C	D	G	H	C	D	G	H								Std.	6.0"	
DN15	mm	11.15	108.00	46.00	333.00	341.90	304.00	367.00	181.40	190.30	152.40	215.40	150.00	89.00	5.50	3/8" UNF	7.20 (F03)	36.00	M5X7	2.40	2.10	7
1/2"	inch	0.44	4.25	1.81	13.11	13.46	11.97	14.45	7.14	7.49	6.00	8.48	5.91	3.50	0.22		0.28	1.42		5.30	4.60	8
DN20	mm	14.30	117.00	49.35	335.40	344.30	304.00	369.40	183.80	192.70	152.40	217.80	150.00	98.00	5.50	3/8" UNF	7.20 (F03)	36.00	M5X7	3.00	2.70	10
3/4"	inch	0.56	4.61	1.94	13.20	13.56	11.97	14.54	7.24	7.59	6.00	8.57	5.91	3.86	0.22		0.28	1.42		6.60	5.90	12
DN25	mm	20.60	127.00	57.00	342.15	359.65	304.00	387.00	190.55	208.05	152.40	235.40	187.00	108.00	7.54	7/16" UNF	7.20 (F04)	42.00	M5X7	4.20	3.85	27
1"	inch	0.81	5.00	2.24	13.47	14.16	11.97	15.24	7.50	8.19	6.00	9.27	7.36	4.25	0.30		0.28	1.65		9.30	8.50	32
DN40	mm	31.80	165.00	62.40	347.55	377.05	304.00	404.00	195.95	225.45	152.40	252.40	237.00	127.00	8.71	9/16" UNF	8.00 (F05)	50.00	M6X8	7.00	6.43	70
1 1/2"	inch	1.25	6.50	2.46	13.68	14.84	11.97	15.91	7.71	8.88	6.00	9.94	9.33	5.00	0.34		0.31	1.97		15.40	14.20	82
DN50	mm	38.10	178.00	68.00	352.25	381.75	304.00	406.00	200.65	230.15	152.40	254.40	237.00	152.00	8.71	9/16" UNF	8.50 (F05)	50.00	M6X8	9.10	8.53	103
2"	inch	1.50	7.01	2.68	13.87	15.03	11.97	15.98	7.90	9.06	6.00	10.02	9.33	5.98	0.34		0.33	1.97		20.10	18.80	120

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	Plug	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
6A***	Support ring	A479 316L, B574 N06022	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2

⁽¹⁾ C31C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C31C design.

Item	Description	Material specifications	Qty.
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1
13	Locking clip	A167 304	1
14	Handle	A240 430	1
15	Serrated washer	A240 410	1
16	Handle nut	EN3506-2 A4-80, A194 Gr 8M	1
17	Sleeve	PVC	1
20	Anti-static spring	A313 302	1
21	Anti-static plunger	A479 304	1
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	A193 B8, EN3506-1 A2-70	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

*** Only with graphite body seal

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

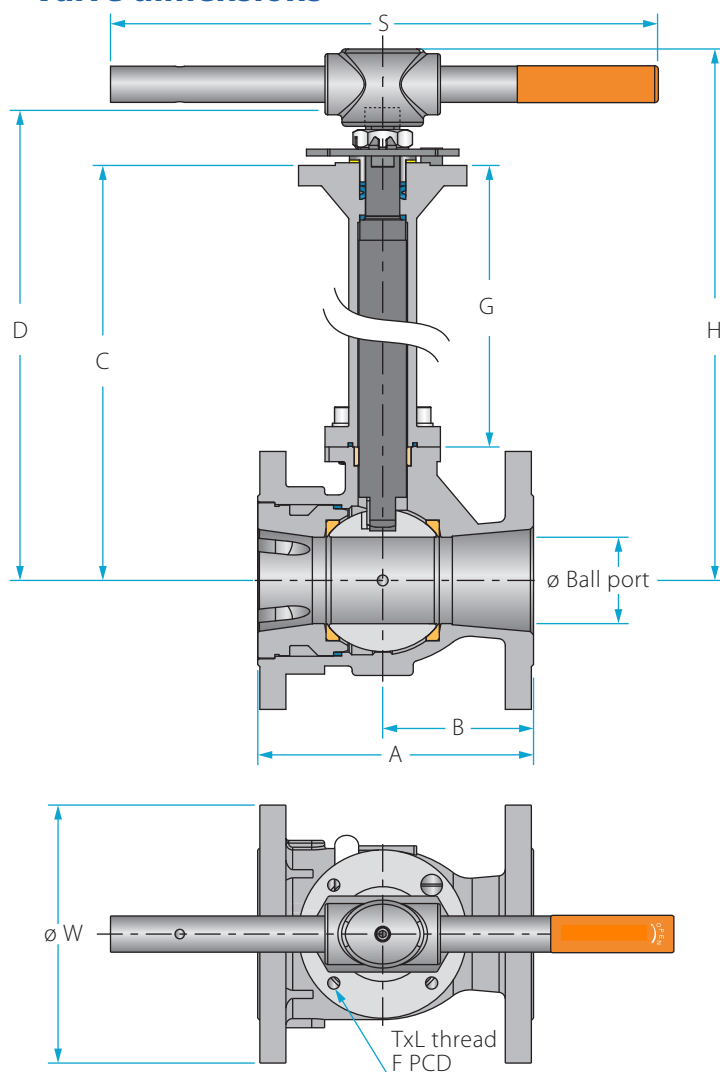
High pressure

Multiport

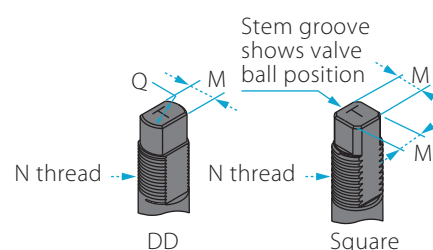
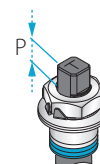
Ordering code system

Size 3"- 8" | DN80-DN200 | ANSI Class 150 | C31W/C31C⁽¹⁾ Series

Valve dimensions



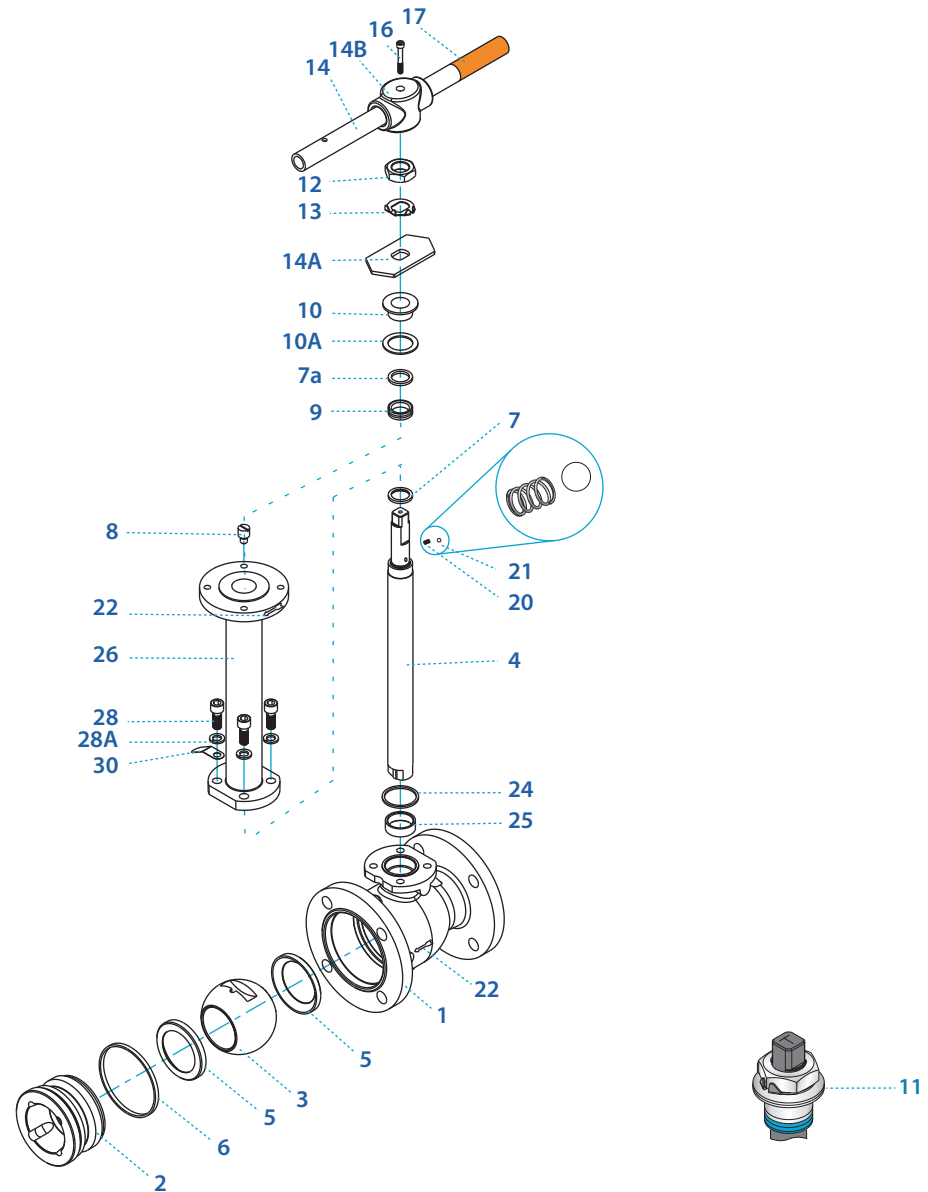
Preparation for actuation



Std. port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	MDD	N	P	Q	F		TxL	Weight kg/lb		Kv
					C	D	G	H	C	D	G	H											Std.	6.0"	Cv
DN80	mm	63.8	282.00	92.10	463.30	509.90	343.50	550.50	272.20	318.80	152.40	359.40	610.00	210.00	18.92	15.90	1"-14	16.70	22.70	(F10)	102.00	M10X15	29.00	27.80	385
3"	inch	2.51	11.10	3.63	18.24	20.07	13.52	21.67	10.72	12.55	6.00	14.15	24.02	8.27	0.74	0.63	UNS-2A	0.66	0.89		4.02		63.90	61.30	450
DN100	mm	82.80	304.80	101.60	457.60	504.20	343.50	543.40	266.50	313.10	152.40	352.30	610.00	254.00	18.92	15.90	1"-14	16.70	22.70	(F10)	102.00	M10X15	44.60	43.40	615
4"	inch	3.26	12.00	4.00	18.02	19.85	13.52	21.39	10.49	12.33	6.00	13.87	24.02	10.00	0.74	0.63	UNS-2A	0.66	0.89		4.02		98.30	95.70	720
DN150	mm	111.3	403.20	107.80	561.00	630.50	385.00	-	328.40	397.90	152.40	-	-	318.00	28.45	23.75	1½"-12	26.20	35.20	(F12)	125.00	M12X15	92.00	89.00	872
6"	inch	4.38	15.87	4.24	22.09	24.82	15.16	-	12.93	15.67	6.00	-	-	12.52	1.12	0.94	UNS-2A	1.03	1.39		4.92		202.40	195.80	1020
DN200	mm	144.40	419.10	142.25	570.10	639.60	385.00	-	337.50	407.00	152.40	-	-	381.00	28.45	23.75	1½"-12	26.20	35.20	(F12)	125.00	M12X15	118.60	114.50	1607
8"	inch	5.69	16.50	5.60	22.44	25.18	15.16	-	13.29	16.02	6.00	-	-	15.00	1.12	0.94	UNS-2A	1.03	1.39		4.92		261.40	252.40	1880

⁽¹⁾ Due to high valve torque, pipe handle cannot be used. A manual gear or automation means should be used to operate the valve.

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 8M	1
13	Tab lock washer	A240 304	1

⁽¹⁾ C73C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C73C design.

Item	Description	Material specifications	Qty.
14***	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN3506-1 A2-70, A193 B8	8-10
18A	Spring washer	DIN 127 A2	8-10
20	Anti-static spring	A313 302	2
21	Anti-static plunger	A479 304	2
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

*** Gear operator should be used for size 6" DN150 and above (handle components are not included)

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

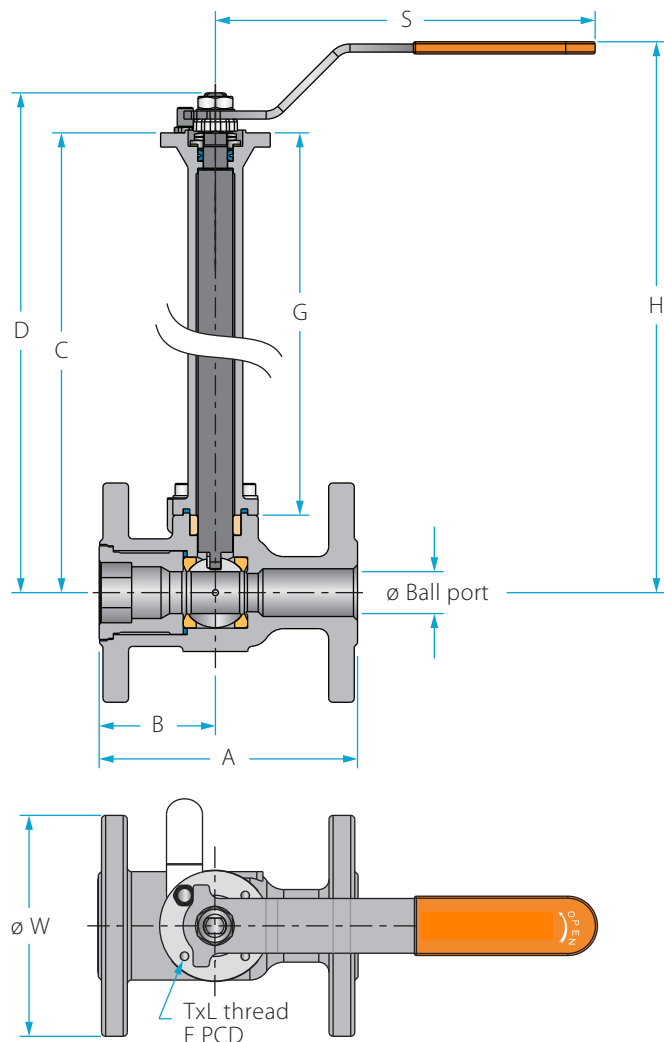
High
pressure

Multiport

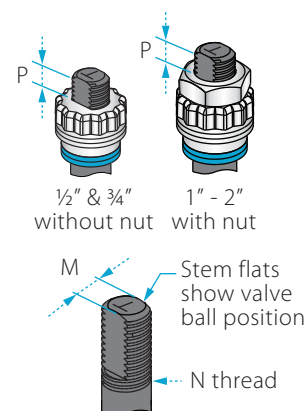
Ordering
code system

Size 1/2"- 2" | DN15-DN50 | ANSI Class 300 | C32W/C32C⁽¹⁾ Series

Valve dimensions

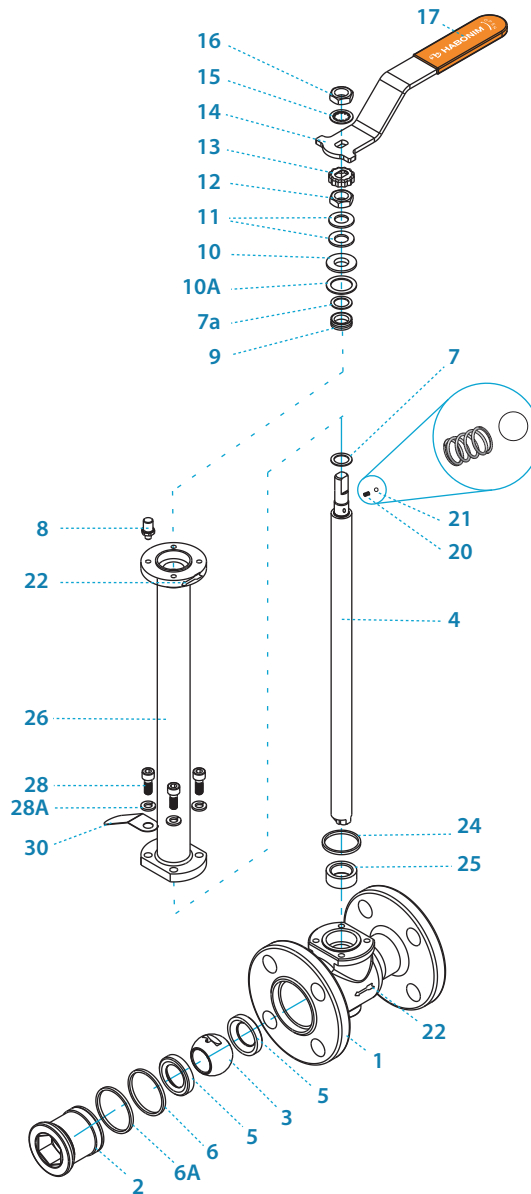


Preparation for actuation



Std. port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	N	P	F	TxL	Weight kg/lb		Kv	
					C	D	G	H	C	D	G	H								Std.	6.0"	Cv	
DN15	mm	11.15	140.00	46.00	333.00	341.90	304.00	367.00	181.40	190.30	152.40	215.40	150.00	95.00	5.50	3/8" UNF	7.20	(F03)	36.00	M5X7	3.10	2.80	7
1/2"	inch	0.44	5.51	1.81	13.11	13.46	11.97	14.45	7.14	7.49	6.00	8.48	5.91	3.74	0.22		0.28		1.42			6.80	6.10
DN20	mm	14.30	152.40	49.33	335.40	344.30	304.00	369.40	183.80	192.70	152.40	217.80	150.00	117.00	5.50	3/8" UNF	7.20	(F03)	36.00	M5X7	4.00	3.70	10
3/4"	inch	0.56	6.00	1.94	13.20	13.56	11.97	14.54	7.24	7.59	6.00	8.57	5.91	4.61	0.22		0.28		1.42			8.80	8.10
DN25	mm	20.60	165.00	57.00	342.05	359.55	304.00	387.00	190.55	208.05	152.40	235.40	187.00	124.00	7.54	7/16" UNF	7.20	(F04)	42.00	M5X7	5.50	5.20	27
1"	inch	0.81	6.50	2.24	13.47	14.16	11.97	15.24	7.50	8.19	6.00	9.27	7.36	4.88	0.30		0.28		1.65			12.10	11.40
DN40	mm	31.80	190.50	62.30	347.45	376.95	304.00	404.00	195.95	225.45	152.40	252.40	237.00	156.00	8.71	9/16" UNF	8.00	(F05)	50.00	M6X8	10.00	9.60	70
1 1/2"	inch	1.25	7.50	2.45	13.68	14.84	11.97	15.91	7.71	8.88	6.00	9.94	9.33	6.14	0.34		0.31		1.97			22.50	21.20
DN50	mm	38.10	216.00	67.80	352.25	381.8	304.00	408.00	200.65	230.15	152.40	256.40	237.00	165.00	8.71	9/16" UNF	8.50	(F05)	50.00	M6X8	12.10	11.70	103
2"	inch	1.50	8.50	2.67	13.87	15.0	11.97	16.06	7.90	9.06	6.00	10.09	9.33	6.50	0.34		0.33		1.97			26.00	24.80

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	Plug	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
6A***	Support ring	A479 316L, B574 N06022	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2

⁽¹⁾ C32C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C32C design.

Item	Description	Material specifications	Qty.
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1
13	Locking clip	A167 304	1
14	Handle	A240 430	1
15	Serrated washer	A240 410	1
16	Handle nut	EN3506-2 A4-80, A194 Gr 8M	1
17	Sleeve	PVC	1
20	Anti-static spring	A313 302	1
21	Anti-static plunger	A479 304	1
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	A193 B8, EN3506-1 A2-70	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

*** Only with graphite body seal

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

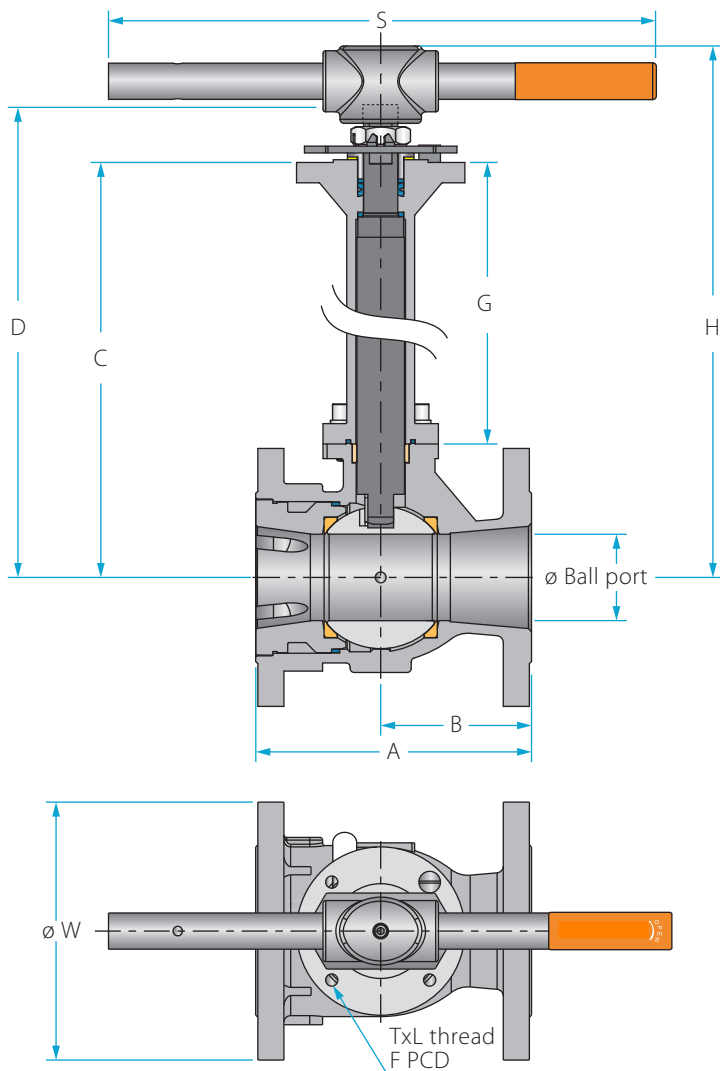
High
pressure

Multiport

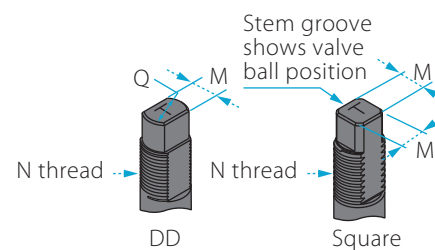
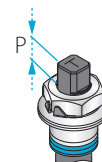
Ordering
code system

Size 3"- 8" | DN80-DN200 | ANSI Class 300 | C32W/C32C⁽¹⁾ Series

Valve dimensions



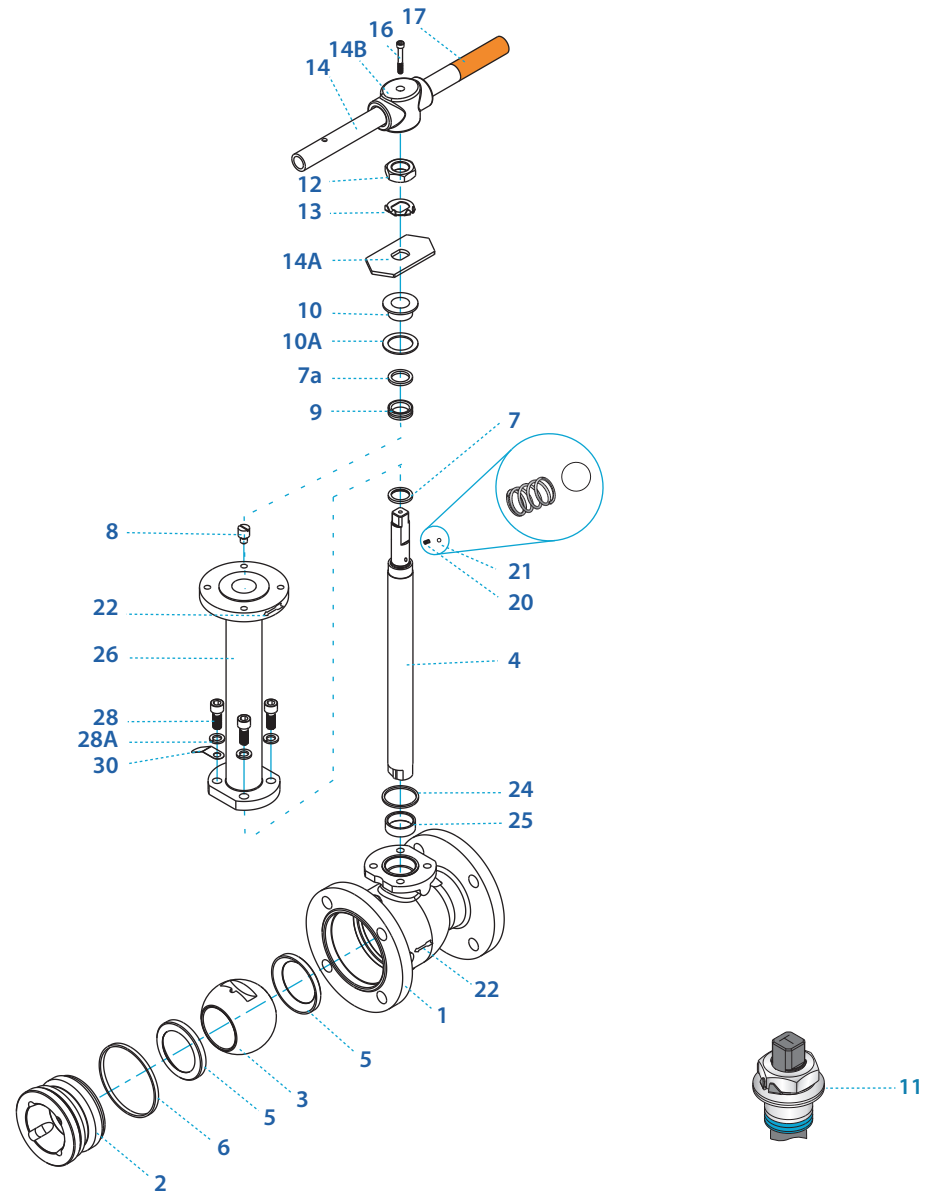
Preparation for actuation



Std. port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	MDD	N	P	Q	F	TxL	Weight kg/lb			Kv
					C	D	G	H	C	D	G	H										Std.	6.0"	Cv	
DN80	mm	63.8	282.00	92.10	463.30	509.90	343.50	550.50	272.20	318.80	152.40	359.40	610.00	210.00	18.92	15.90	1"-14	16.70	22.70	(F10)	102.00	M10X15	29.00	27.80	385
3"	inch	2.51	11.10	3.63	18.24	20.07	13.52	21.67	10.72	12.55	6.00	14.15	24.02	8.27	0.74	0.63	UNS-2A	0.66	0.89		4.02		63.90	61.30	450
DN100	mm	82.80	304.80	101.60	457.60	504.20	343.50	543.40	266.50	313.10	152.40	352.30	610.00	254.00	18.92	15.90	1"-14	16.70	22.70	(F10)	102.00	M10X15	44.60	43.40	615
4"	inch	3.26	12.00	4.00	18.02	19.85	13.52	21.39	10.49	12.33	6.00	13.87	24.02	10.00	0.74	0.63	UNS-2A	0.66	0.89		4.02		98.30	95.70	720
DN150	mm	111.3	403.20	107.80	561.00	630.50	385.00	-	328.40	397.90	152.40	-	-	318.00	28.45	23.75	1-1/2"	26.20	35.20	(F12)	125.00	M12X15	92.00	89.00	872
6"	inch	4.38	15.87	4.24	22.09	24.82	15.16	-	12.93	15.67	6.00	-	-	12.52	1.12	0.94	UNF-2A	1.03	1.39		4.92		202.40	195.80	1020
DN200	mm	144.40	419.10	142.25	570.10	639.60	385.00	-	337.50	407.00	152.40	-	-	381.00	28.45	23.75	1-1/2"	26.20	35.20	(F12)	125.00	M12X15	118.60	114.50	1607
8"	inch	5.69	16.50	5.60	22.44	25.18	15.16	-	13.29	16.02	6.00	-	-	15.00	1.12	0.94	UNF-2A	1.03	1.39		4.92		261.40	252.40	1880

⁽¹⁾ Due to high valve torque, pipe handle cannot be used. A manual gear or automation means should be used to operate the valve.

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	Plug	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1
13	Tab lock washer	A240 304	1

Item	Description	Material specifications	Qty.
14***	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
20	Anti-static spring	A313 302	2
21	Anti-static plunger	A479 304	2
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	A193 B8, EN3506-1 A2-70	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

*** Gear operator should be used for size 6" DN150 and above (handle components are not included)

⁽¹⁾ C32C series is Habonim's standard valve design without the HermetiX™ stem seal construction.⁽²⁾ This material can only be used as part of the C32C design.

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

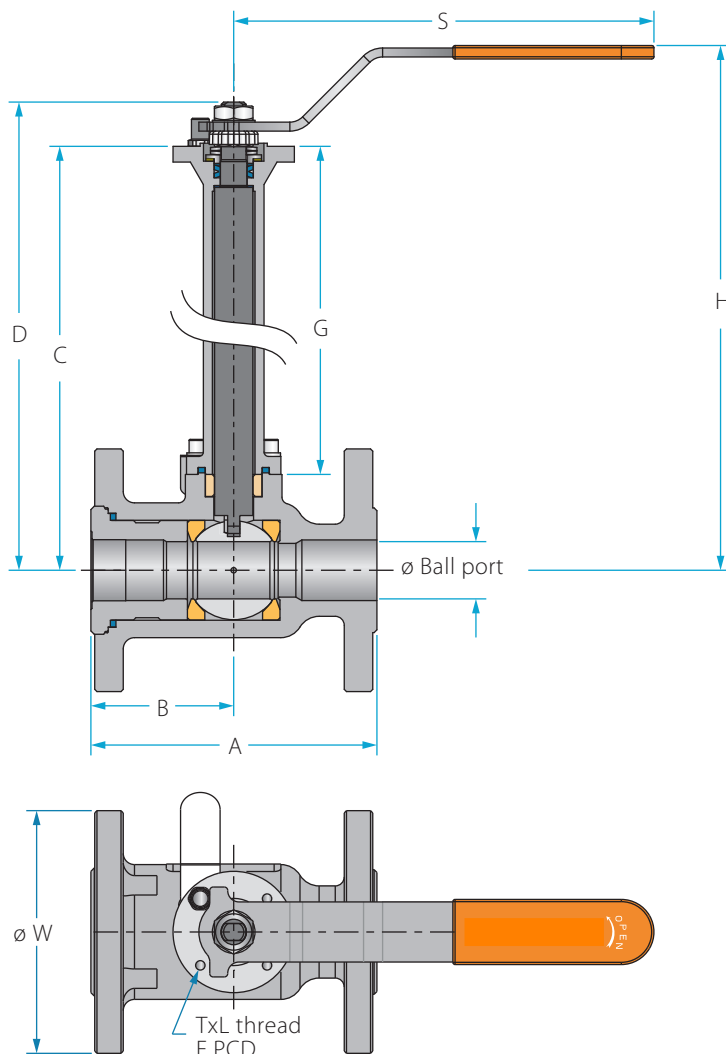
High
pressure

Multiport

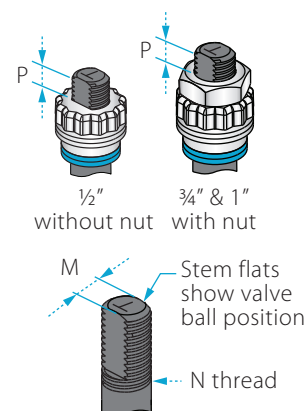
Ordering
code system

Size 1/2"-1" | DN15-DN25 | ANSI Class 150 | C73W/C73C⁽¹⁾ Series

Valve dimensions

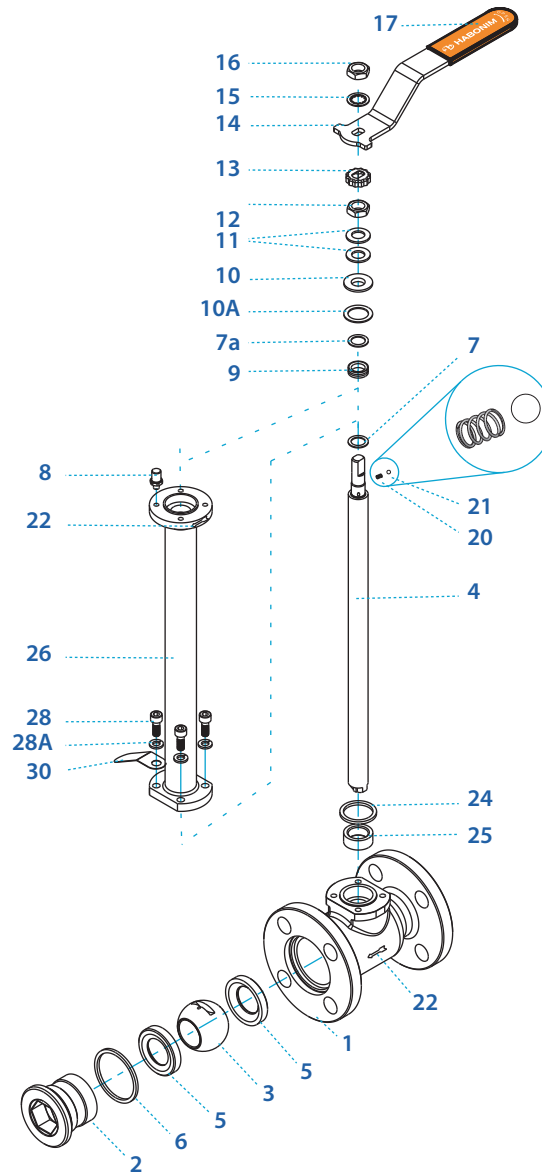


Preparation for actuation



Full Port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	N	P	F	TxL	Weight kg/lb		Kv
					C	D	G	H	C	D	G	H								Std.	6.0"	
DN15	mm	14.30	108.20	48.20	335.40	344.30	304.00	400.00	183.80	192.70	152.40	248.40	150.00	89.00	5.50	3/8" UNF	7.20 (F03)	36.00	M5X7	2.60	2.28	26
1/2"	inch	0.56	4.26	1.90	13.20	13.56	11.97	15.75	7.24	7.59	6.00	9.78	5.91	3.50	0.22		0.28	1.42		5.73	5.03	30
DN20	mm	20.60	117.00	59.00	342.15	359.65	304.00	406.80	190.55	208.05	152.40	255.20	187.00	98.00	7.54	7/16" UNF	7.20 (F04)	42.00	M5X7	3.80	3.45	43
3/4"	inch	0.81	4.61	2.32	13.47	14.16	11.97	16.02	7.50	8.19	6.00	10.05	7.36	3.86	0.30		0.28	1.65		8.38	7.60	50
DN25	mm	25.40	127.00	64.00	346.65	364.15	304.00	411.30	195.05	212.55	152.40	259.70	187.00	108.00	7.54	7/16" UNF	7.20 (F04)	42.00	M5X7	4.60	4.25	85
1"	inch	1.00	5.00	2.52	13.65	14.34	11.97	16.19	7.68	8.37	6.00	10.22	7.36	4.25	0.30		0.28	1.65		10.14	9.37	105

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	Plug	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	2
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1

⁽¹⁾ C73C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C73C design.

Item	Description	Material specifications	Qty.
13	Locking clip	A167 304	1
14	Handle	A240 430	1
15	Serrated washer	A240 410	1
16	Handle nut	EN3506-2 A4-80, A194 Gr 8M	1
17	Sleeve	PVC	1
20	Anti-static spring	A313 302	1
21	Anti-static plunger	A479 304	1
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	A193 B8, EN3506-1 A2-70	4
28A	Spring washer	DIN 912 A2-70	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

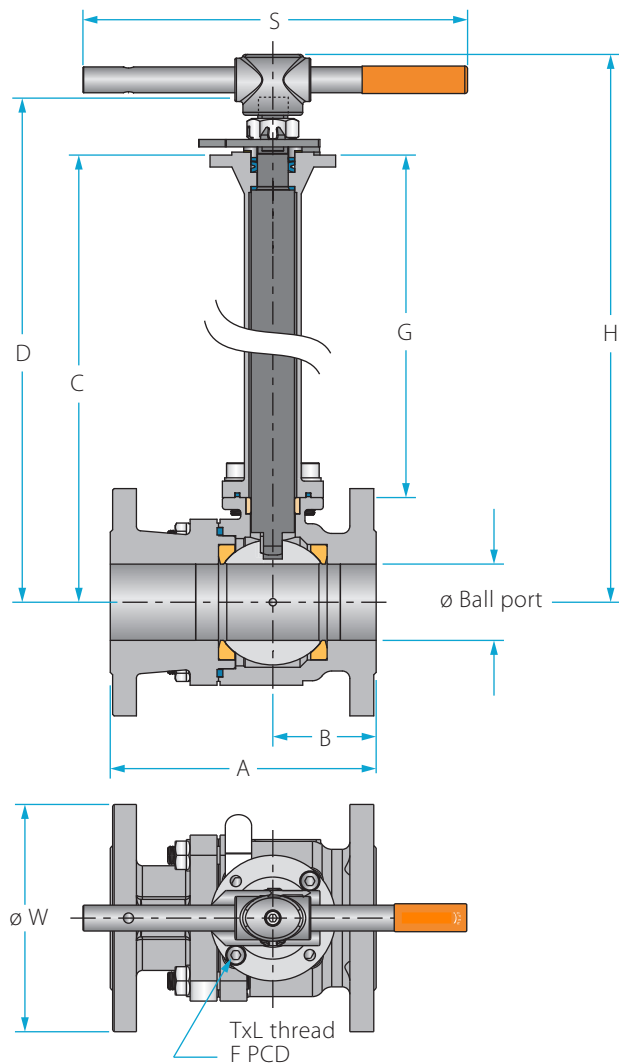
High
pressure

Multiport

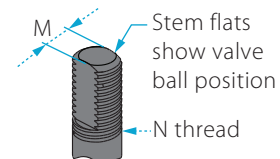
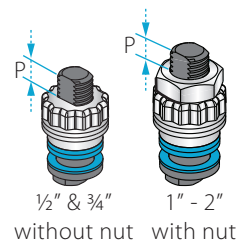
Ordering
code system

Size 1½"- 2" | DN40-DN50 | ANSI Class 150 | **C73W/C73C⁽¹⁾** Series

Valve dimensions

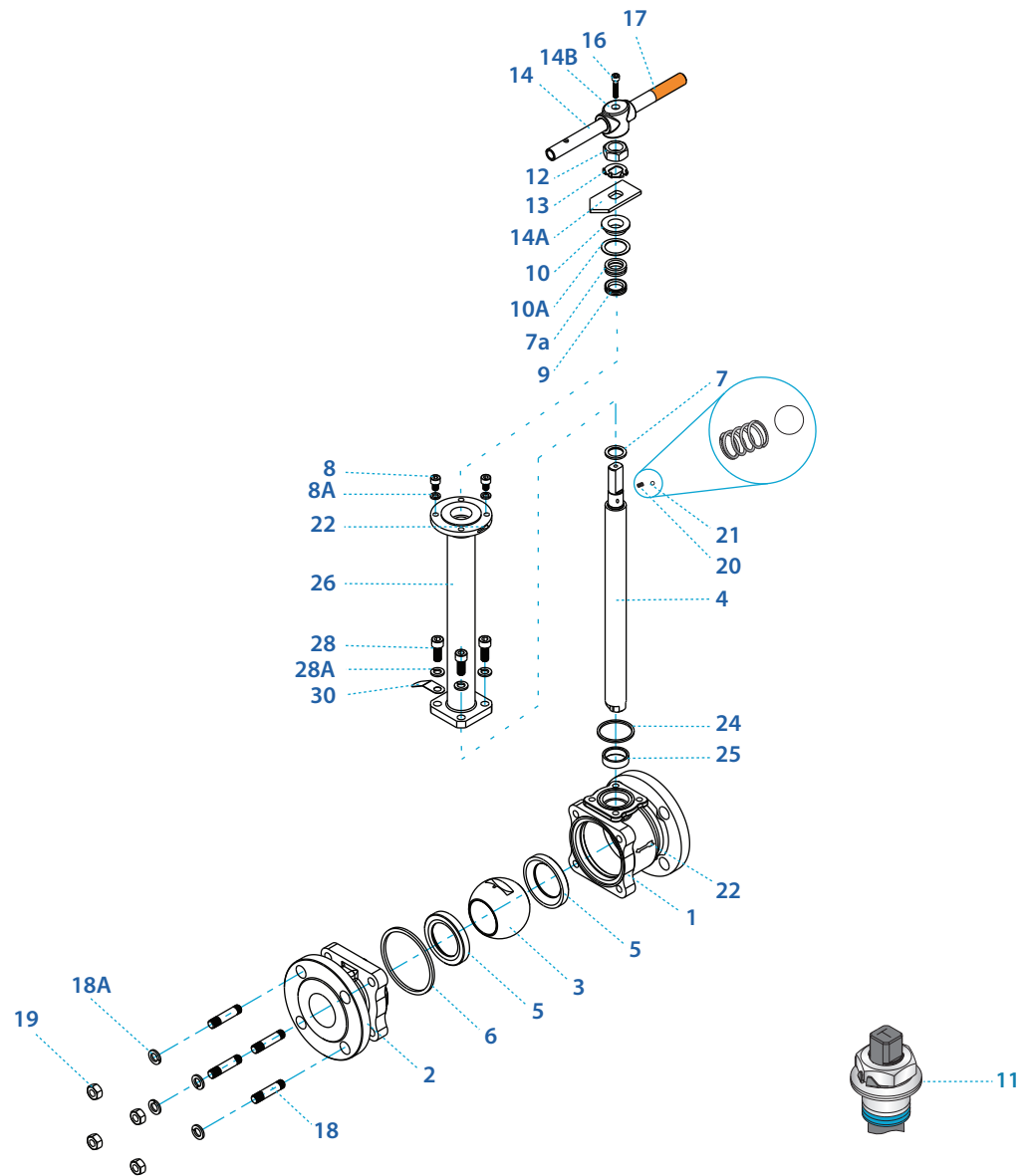


Preparation for actuation



Full Port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	N	P	F	TxL	Weight kg/lb			Kv
					C	D	G	H	C	D	G	H								Std.	6.0"	Cv	
DN40	mm	38.10	165.00	68.00	352.3	381.8	304.00	406.60	200.65	230.15	152.40	255.00	237.00	127.00	8.71	9/16" UNF	8.50 (F05)	50.00	M6X8	8.50	7.93	214	
1½"	inch	1.50	6.50	2.68	13.87	15.0	11.97	16.01	7.90	9.06	6.00	10.04	9.33	5.00	0.34		0.33	1.97		18.73	17.48	250	
DN50	mm	51.00	178.00	69.00	397.50	439.10	327.50	430.35	222.40	264.00	152.40	255.25	400.00	152.00	13.90	M20x2.5	13.50 (F07)	70.00	M8X8	16.00	14.50	410	
2"	inch	2.01	7.01	2.72	15.65	17.29	12.89	16.94	8.76	10.39	6.00	10.05	15.75	5.98	0.55		0.53	2.76		35.20	31.90	480	

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop bolt	A582 303	2
8A	Spring washer	DIN 127 A2	2
9*	Stem seal	TFM, Graphite	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
12	Disc spring	A693 Gr. 631 17-7PH	2
13	Stem nut	EN3506-2 A4-80, A194 8M	1
13	Tab lock washer	A240 304	1

⁽¹⁾ C73C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C73C design.

Item	Description	Material specifications	Qty.
14	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	4
18A	Spring washer	DIN 127 A2	4
19	Body Nut	EN3506-2 A4-80, A194 8M	4
20	Anti-static spring	A313 302	1
21	Anti-static plunger	A479 304	1
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

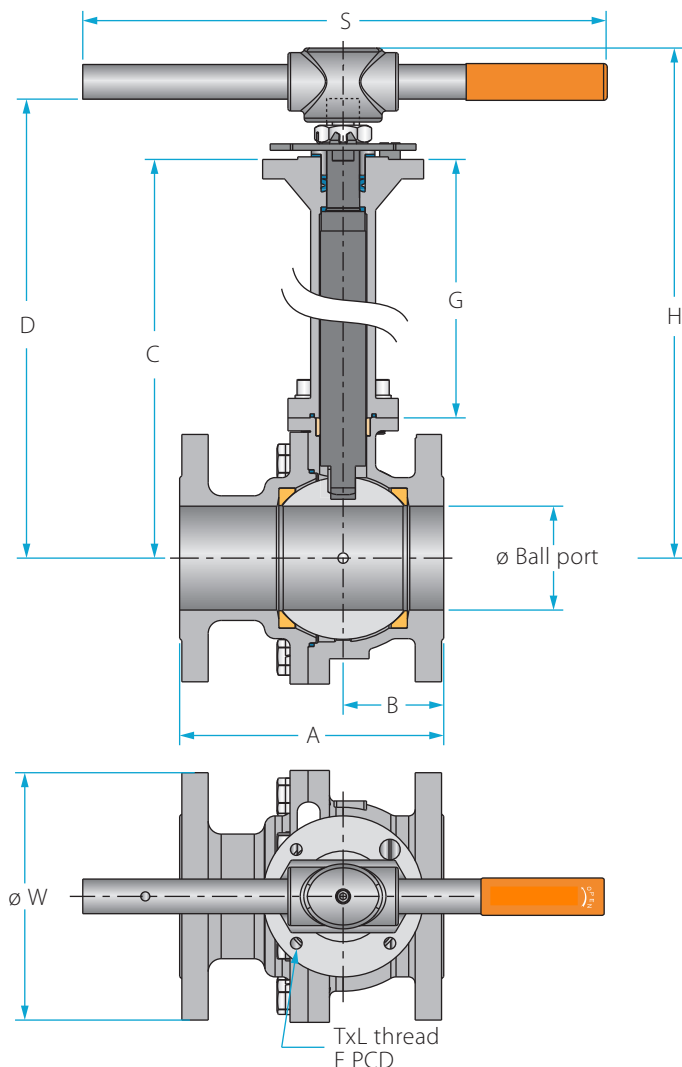
High
pressure

Multiport

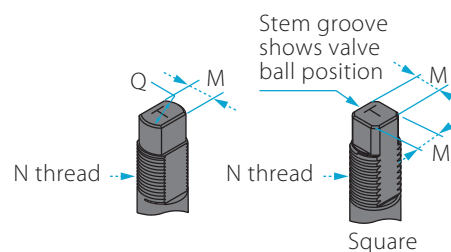
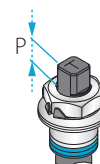
Ordering
code system

Size 3"- 8" | DN80-DN200 | ANSI Class 150 | C73W/C73C⁽¹⁾ Series

Valve dimensions



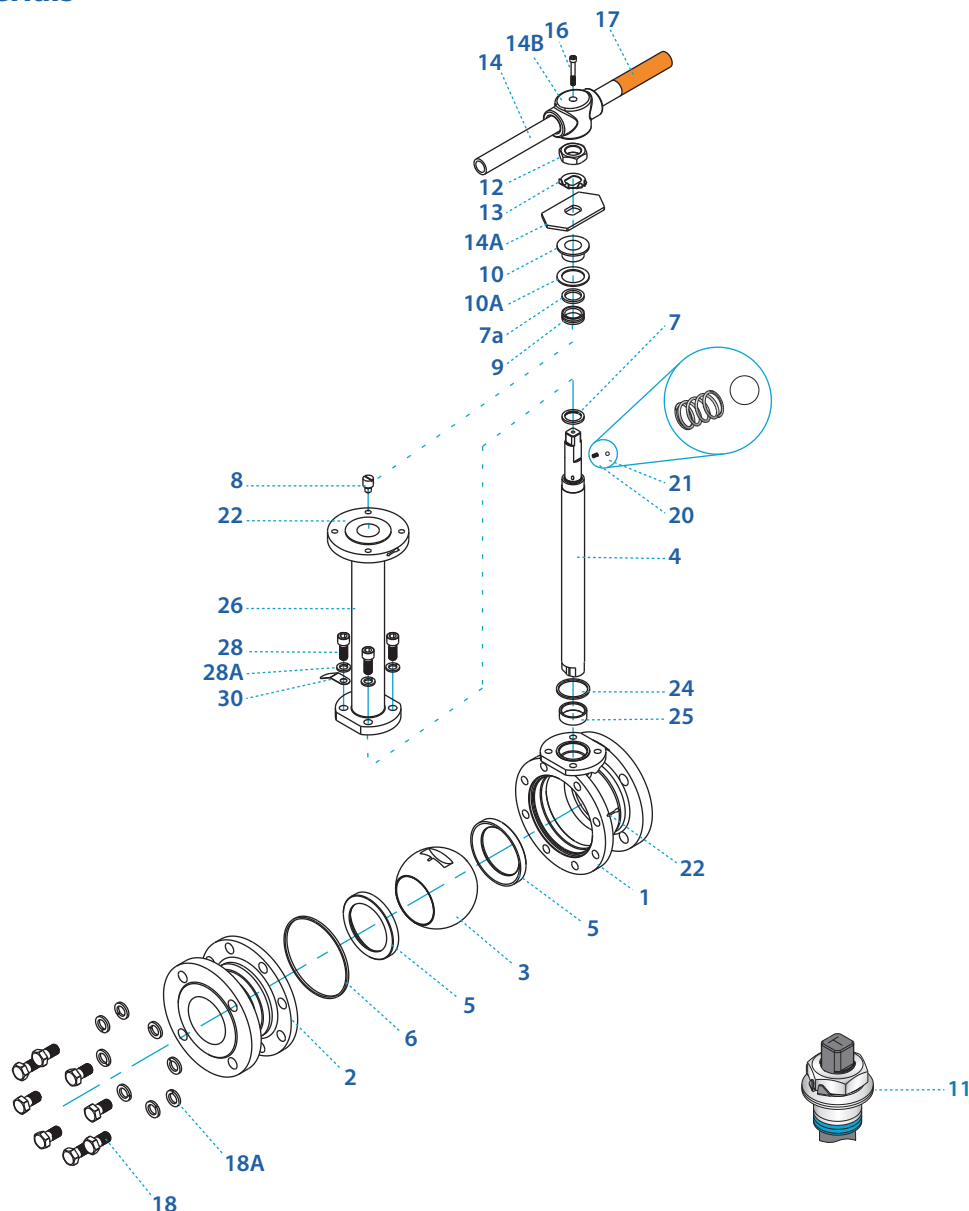
Preparation for actuation



Full Port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	M DD	N	P	Q	F	TxL	Weight kg/lb		Kv	
					C	D	G	H	C	D	G	H										Std.	6.0"	Cv	
DN80	mm	80.00	203.00	77.50	441.50	488.10	343.50	538.60	250.40	297.00	152.40	347.50	610.00	190.50	18.90	15.90	1"-14	16.70	22.70	(F10)	102.00	M10X15	27.00	25.00	1111
3"	inch	3.15	7.99	3.05	17.38	19.22	13.52	21.20	9.86	11.69	6.00	13.68	24.02	7.50	0.74	0.63	UNS-2A	0.66	0.89		4.02		59.40	41.44	1300
DN100	mm	100.00	229.00	84.50	467.50	514.10	343.50	566.60	276.40	323.00	152.40	375.50	610.00	230.00	18.90	15.90	1"-14	16.70	22.70	(F10)	102.00	M10X15	44.00	42.00	2051
4"	inch	3.94	9.02	3.33	18.41	20.24	13.52	22.31	10.88	12.72	6.00	14.78	24.02	9.06	0.74	0.63	UNS-2A	0.66	0.89		4.02		67.22	64.58	2379
DN150	mm	150.00	394.00	163.50	564.00	633.50	385.00	-	331.40	400.90	152.40	-	-	279.40	28.45	23.75	1-1/2"-12 UNF-2A	26.20	35.20	(F12)	125.00	M12X15	97.00	93.50	4615
6"	inch	5.91	15.51	6.44	22.20	24.94	15.16	-	13.05	15.78	6.00	-	-	11.00	1.12	0.94		1.03	1.39		4.92		154.28	145.24	5400
DN200	mm	200.00	457.00	242.60	627.00	627.00	385.00	-	394.40	463.90	152.40	-	-	343.00	35.90	35.90	2" UN-2A	40.00	46.50	(F14)	140.00	M16X20	210.00	203.00	7949
8"	inch	7.87	17.99	9.55	24.69	24.69	15.16	-	15.53	18.26	6.00	-	-	13.50	1.41	1.41		1.57	1.83		5.51		231.42	222.38	9300

⁽¹⁾ Due to high valve torque, pipe handle cannot be used. A manual gear or automation means should be used to operate the valve.

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 8M	1
13	Tab lock washer	A240 304	1

⁽¹⁾ C73C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C73C design.

Item	Description	Material specifications	Qty.
14***	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN3506-1 A2-70, A193 B8	8-10
18A	Spring washer	DIN 127 A2	8-10
20	Anti-static spring	A313 302	2
21	Anti-static plunger	A479 304	2
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

*** Gear operator should be used for size 6" DN150 and above (handle components are not included)

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

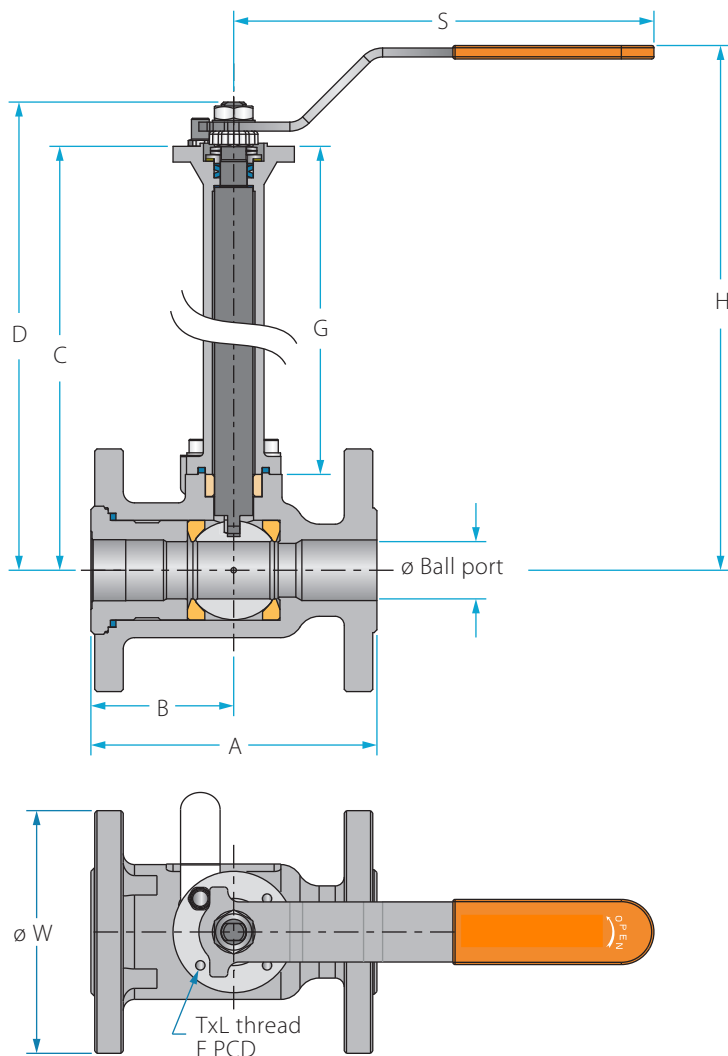
High
pressure

Multiport

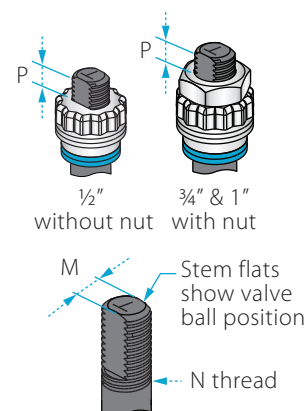
Ordering
code system

Size 1/2"-1" | DN15-DN25 | ANSI Class 300 | C74W/C74C⁽¹⁾ Series

Valve dimensions

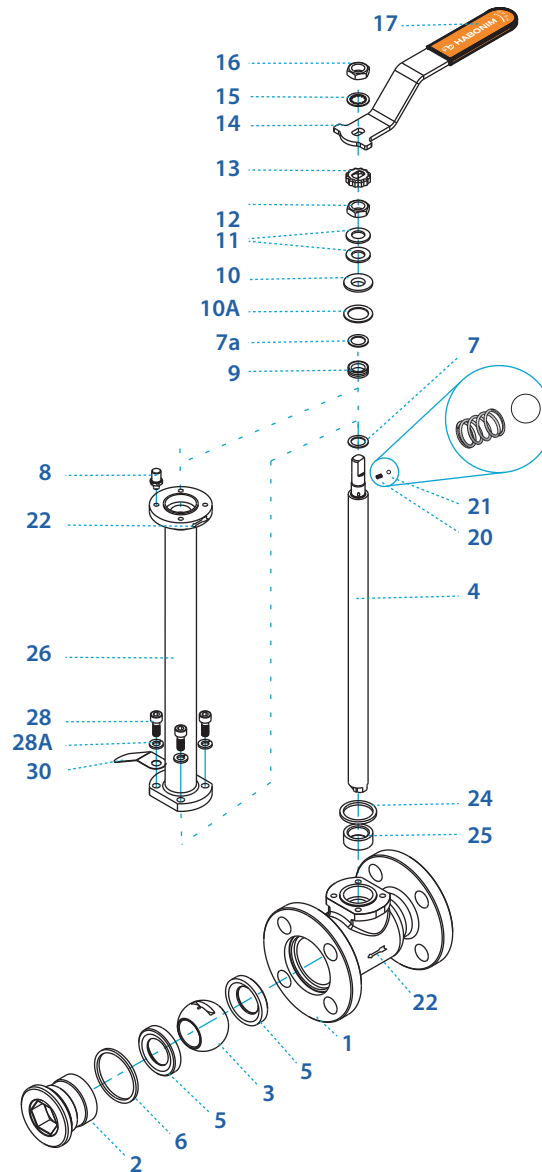


Preparation for actuation



Full Port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	N	P	F	TxL	Weight kg/lb		Kv
					C	D	G	H	C	D	G	H								Std.	6.0"	
DN15	mm	14.30	140.00	48.00	335.40	344.30	304.00	400.00	183.80	192.70	152.40	248.40	150.00	95.00	5.50	3/8" UNF	7.20 (F03)	36.00	M5X7	3.10	2.78	26
1/2"	inch	0.56	5.51	1.89	13.20	13.56	11.97	15.75	7.24	7.59	6.00	9.78	5.91	3.74	0.22		0.28	1.42		6.83	6.13	30
DN20	mm	20.60	153.00	59.00	342.2	359.7	304.00	406.80	190.55	208.05	152.40	255.20	187.00	117.00	7.54	7/16" UNF	7.20 (F04)	42.00	M5X8	4.90	4.55	43
3/4"	inch	0.81	6.02	2.32	13.47	14.16	11.97	16.02	7.50	8.19	6.00	10.05	7.36	4.61	0.30		0.28	1.65		10.80	10.03	50
DN25	mm	25.40	165.00	64.00	346.7	364.15	304.00	411.30	195.05	212.55	152.40	259.70	187.00	124.00	7.54	7/16" UNF	7.20 (F04)	42.00	M5X8	6.00	5.65	85
1"	inch	1.00	6.50	2.52	13.65	14.34	11.97	16.19	7.68	8.37	6.00	10.22	7.36	4.88	0.30		0.28	1.65		13.22	12.45	105

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	Plug	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	2
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1

⁽¹⁾ C74C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C74C design.

Item	Description	Material specifications	Qty.
13	Locking clip	A167 304	1
14	Handle	A240 430	1
15	Serrated washer	A240 410	1
16	Handle nut	EN3506-2 A4-80, A194 Gr 8M	1
17	Sleeve	PVC	1
20	Anti-static spring	A313 302	1
21	Anti-static plunger	A479 304	1
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

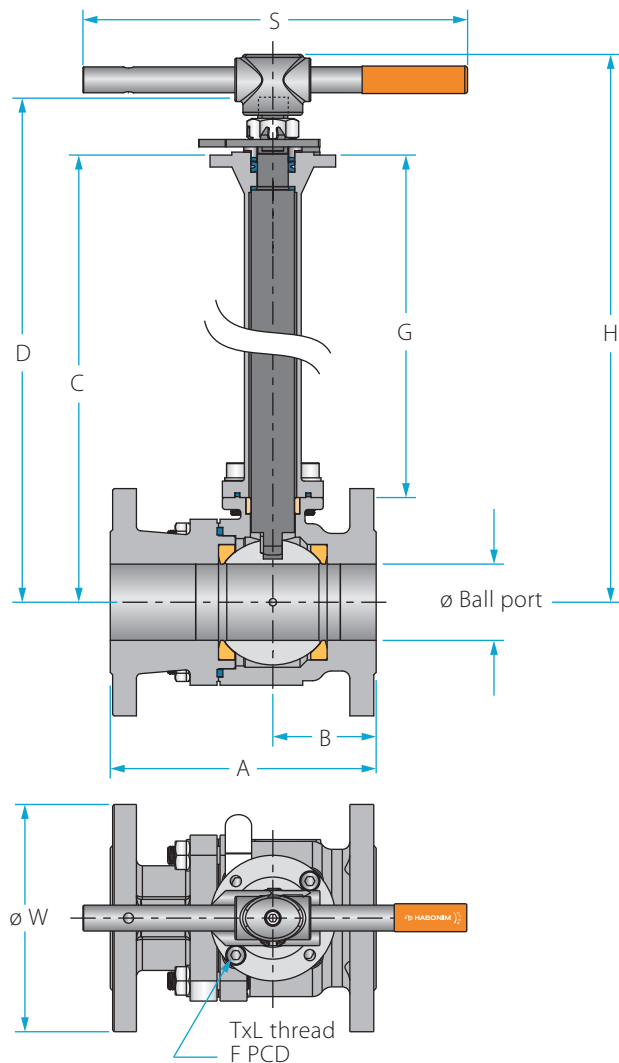
High
pressure

Multiport

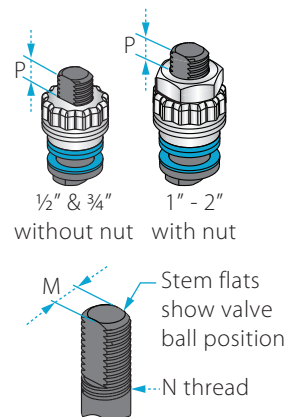
Ordering
code system

Size 1½" - 2" | DN40-DN50 | ANSI Class 300 | **C74W/C74C⁽¹⁾** Series

Valve dimensions

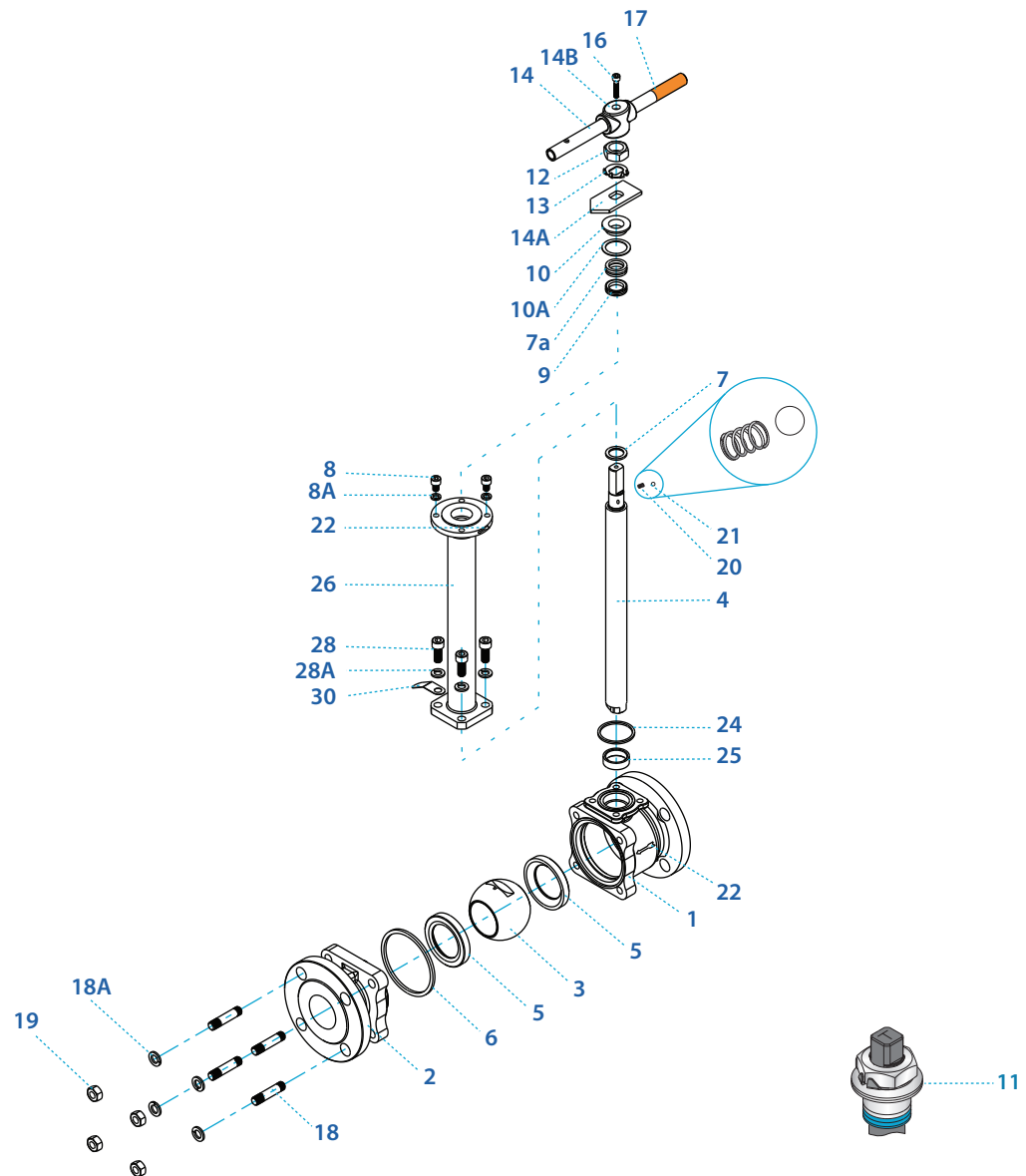


Preparation for actuation



Full Port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	N	P	F	TxL	Weight kg/lb		
					C	D	G	H	C	D	G	H								Std.	6.0"	Cv
DN40	mm	38.10	190.50	110.00	352.25	381.75	304.00	408.00	200.65	230.15	152.40	256.40	237.00	156.00	8.71	9/16" UNF	8.50 (F05)	50.00	M6X8	11.70	11.13	214
1½"	inch	1.50	7.50	4.33	13.87	15.03	11.97	16.06	7.90	9.06	6.00	10.09	9.33	6.14	0.34		0.33	1.97		25.79	24.53	250
DN50	mm	51.00	216.00	141.00	374.00	415.60	304.00	442.60	222.40	264.00	152.40	291.00	400.00	165.00	13.90	M20X2.5	13.50 (F07)	70.00	M8X8	18.00	16.50	410
2"	inch	2.01	8.50	5.55	14.72	16.36	11.97	17.43	8.76	10.39	6.00	11.46	15.75	6.50	0.55		0.53	2.76		39.60	36.30	480

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop bolt	A582 303	2
8A	Spring washer	DIN 127 A2	2
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
12	Disc spring	A693 Gr. 631 17-7PH	2
13	Stem nut	EN3506-2 A4-80, A194 8M	1
13	Tab lock washer	A240 304	1

⁽¹⁾ C74C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C74C design.

Item	Description	Material specifications	Qty.
14	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body stud	EN3506-2 A2-70, A193 B8	4
18A	Spring washer	DIN 127 A2	4
19	Body Nut	EN3506-2 A4-80, A194 8M	4
20	Anti-static spring	A313 302	1
21	Anti-static plunger	A479 304	1
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

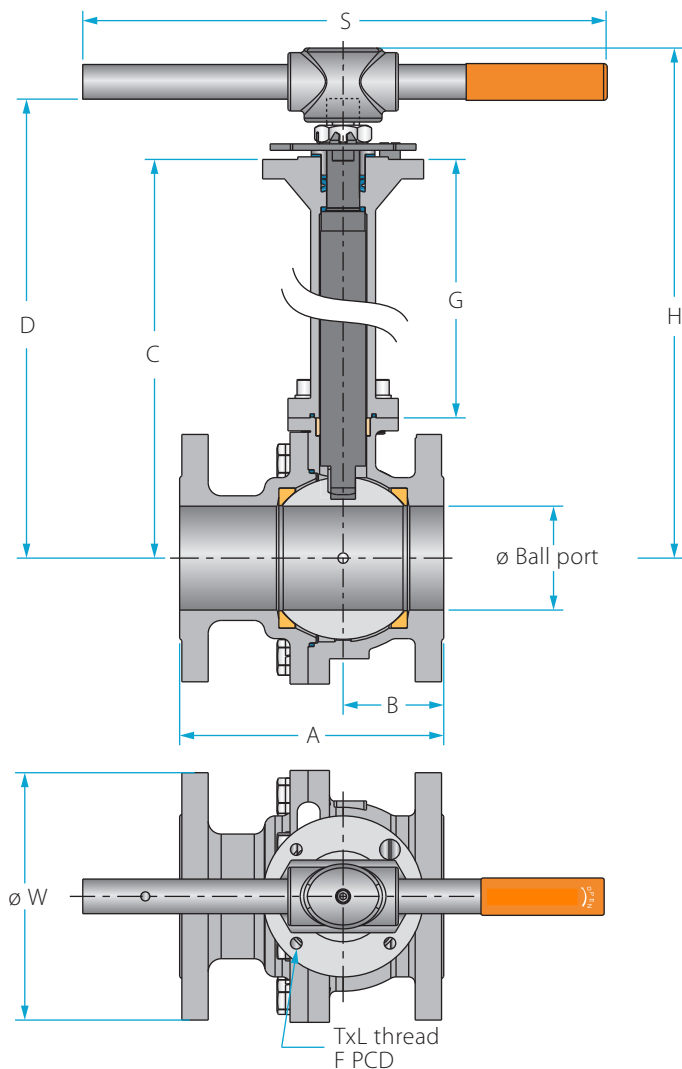
High
pressure

Multiport

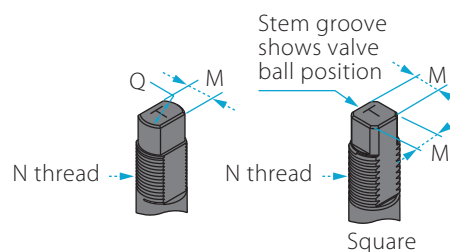
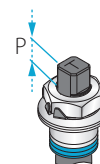
Ordering
code system

Size 3"- 8" | DN80-DN200 | ANSI Class 300 | C74W/C74C⁽¹⁾ Series

Valve dimensions



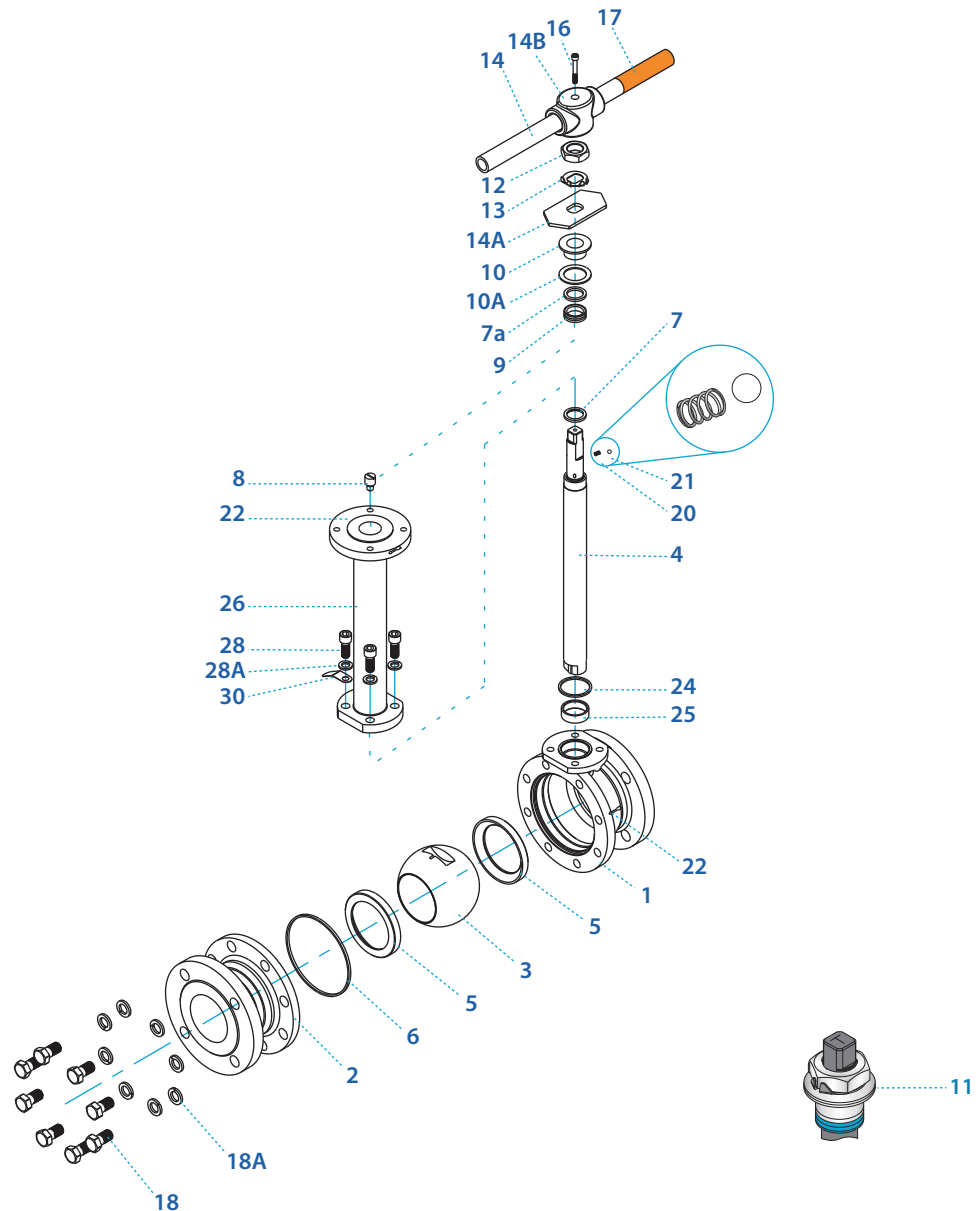
Preparation for actuation



Full Port	Unit	Ball port	A	B	Standard design				6.0" bonnet				S	W	M	M DD	N	P	Q	F	TxL	Weight kg/lb		Kv	
					C	D	G	H	C	D	G	H										Std.	6.0"	Cv	
DN80	mm	80.00	282.00	96.50	451.50	498.10	343.50	538.60	260.40	307.00	152.40	347.50	610.00	210.00	18.90	18.90	1"-14	16.70	22.70	(F10)	102.00	M10X15	31.50	30.30	1111
3"	inch	3.15	11.10	3.80	17.78	19.61	13.52	21.20	10.25	12.09	6.00	13.68	24.02	8.27	0.74	0.74	UNS-2A	0.66	0.89		4.02		69.43	6.78	1300
DN100	mm	100.00	305.00	105.00	467.50	514.10	343.50	554.80	276.40	323.00	152.40	363.70	610.00	254.00	18.90	18.90	1"-14	16.70	22.70	(F10)	102.00	M10X15	45.50	44.30	2051
4"	inch	3.94	12.01	4.13	18.41	20.24	13.52	21.84	10.88	12.72	6.00	14.32	24.02	10.00	0.74	0.74	UNS-2A	0.66	0.89		4.02		100.28	97.64	2400
DN150	mm	150.00	404.00	174.50	564.00	633.50	385.00	-	331.40	400.90	152.40	-	-	328.00	28.45	23.75	1½"-12	26.20	35.20	(F12)	125.00	M12X15	110.00	98.40	4615
6"	inch	5.91	15.91	6.87	22.20	24.94	15.16	-	13.05	15.78	6.00	-	-	12.91	1.12	0.94	UNF-2A	1.03	1.39		4.92		225.91	216.87	5400
DN200	mm	200.00	502.00	241.00	626.00	738.65	385.00	-	393.40	506.05	152.40	-	-	381.00	35.90	35.90	2" UN-2A	40.00	46.50	(F14)	140.00	M16X20	240.00	228.4	7949
8"	inch	7.87	19.76	9.49	24.65	29.08	15.16	-	15.49	19.92	6.00	-	-	15.00	1.41	1.41		1.57	1.83		5.51		528.00	503.39	9300

⁽¹⁾ Due to high valve torque, pipe handle cannot be used. A manual gear or automation means should be used to operate the valve.

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 8M	1
13	Tab lock washer	A240 304	1

⁽¹⁾ C74C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C74C design.

Item	Description	Material specifications	Qty.
14***	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN3506-1 A2-70, A193 B8	8-10
18A	Spring washer	DIN 127 A2	8-10
20	Anti-static spring	A313 302	2
21	Anti-static plunger	A479 304	2
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	A193 B8, DIN 912 A2-70	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

*** Gear operator should be used for size 6" DN150 and above (handle components are not included)

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

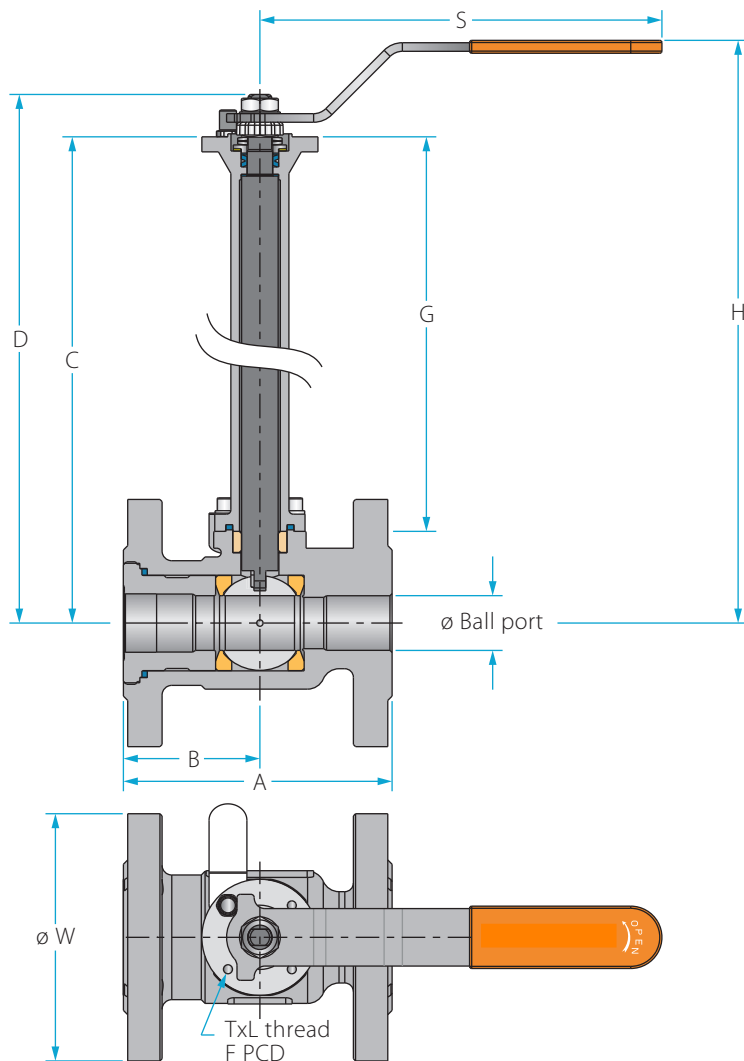
High
pressure

Multiport

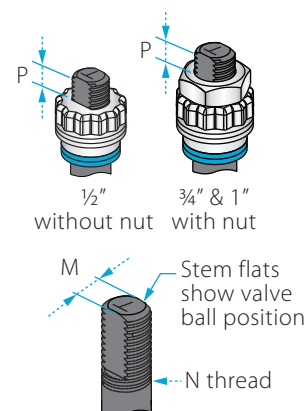
Ordering
code system

Size 1/2" - 1" | DN15-DN25 | DIN PN40 | C78W/C78C⁽¹⁾ Series

Valve dimensions

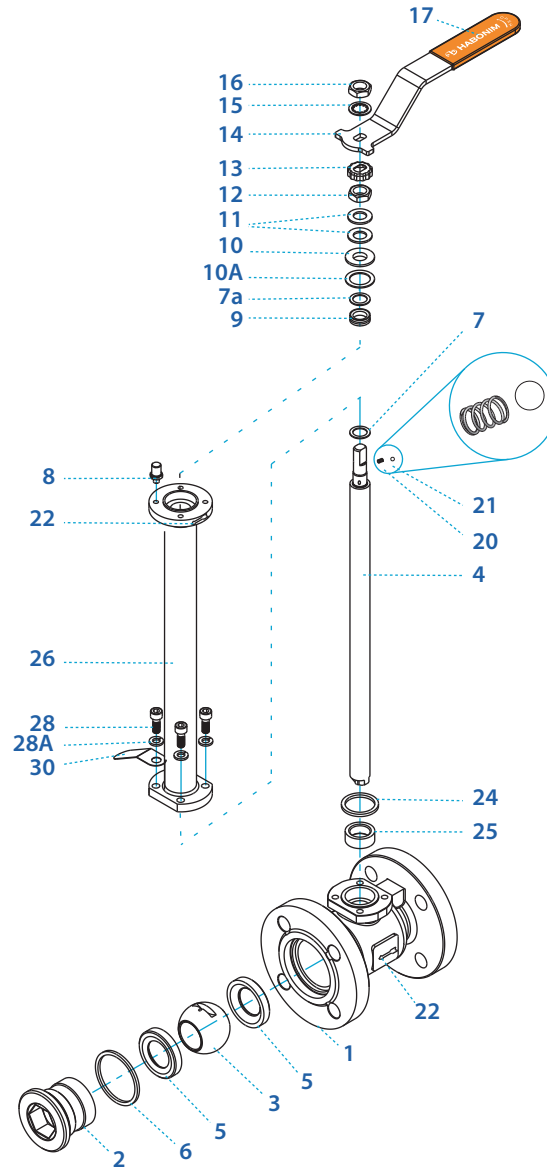


Preparation for actuation



Full Port	Unit	Ball port	A		B	Standard design				6.0" bonnet				S	W	M	N	P	F		TxL	weight kg/lb			Kv
			F1	F4		C	D	G	H	C	D	G	H									F1	F4	Cv	
DN15	mm	80.00	130.00	115.00	48.00	335.40	344.30	304.00	369.60	183.80	192.70	152.40	218.00	150.00	95.00	5.50	3/8" UNF	7.20	(F03)	36.00	M5X7	3.10	3.00	27	
1/2"	inch	3.15	5.12	4.53	1.89	13.20	13.56	11.97	14.55	7.24	7.59	6.00	8.58	15.79	3.74	0.22		0.28		1.42		6.80	6.60	32	
DN20	mm	100.00	150.00	120.00	58.60	342.2	359.7	304.00	383.50	190.55	208.05	152.40	231.90	187.00	105.00	7.50	7/16" UNF	7.20	(F04)	42.00	M5X7	4.50	4.30	46	
3/4"	inch	3.94	5.91	4.72	2.31	13.47	14.16	11.97	15.10	7.50	8.19	6.00	9.13	7.36	4.13	0.30		0.28		1.65		9.80	9.40	54	
DN25	mm	150.00	160.00	125.00	61.50	346.65	364.15	304.00	394.00	195.05	212.55	152.40	242.40	187.00	115.00	7.50	7/16" UNF	7.20	(F04)	42.00	M5X7	5.50	5.30	90	
1"	inch	5.91	6.30	4.92	2.42	13.65	14.34	11.97	15.51	7.68	8.37	6.00	9.54	7.36	4.53	0.30		0.28		1.65		12.00	11.60	105	

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	Plug	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2

⁽¹⁾ C78C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C78C design.

Item	Description	Material specifications	Qty.
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1
13	Locking clip	A167 304	1
14	Handle	A240 430	1
15	Serrated washer	A240 410	1
16	Handle nut	EN3506-2 A4-80, A194 Gr 8M	1
17	Sleeve	PVC	1
20	Anti-static spring	A313 302	1
21	Anti-static plunger	A479 304	1
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 912 A2-70	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

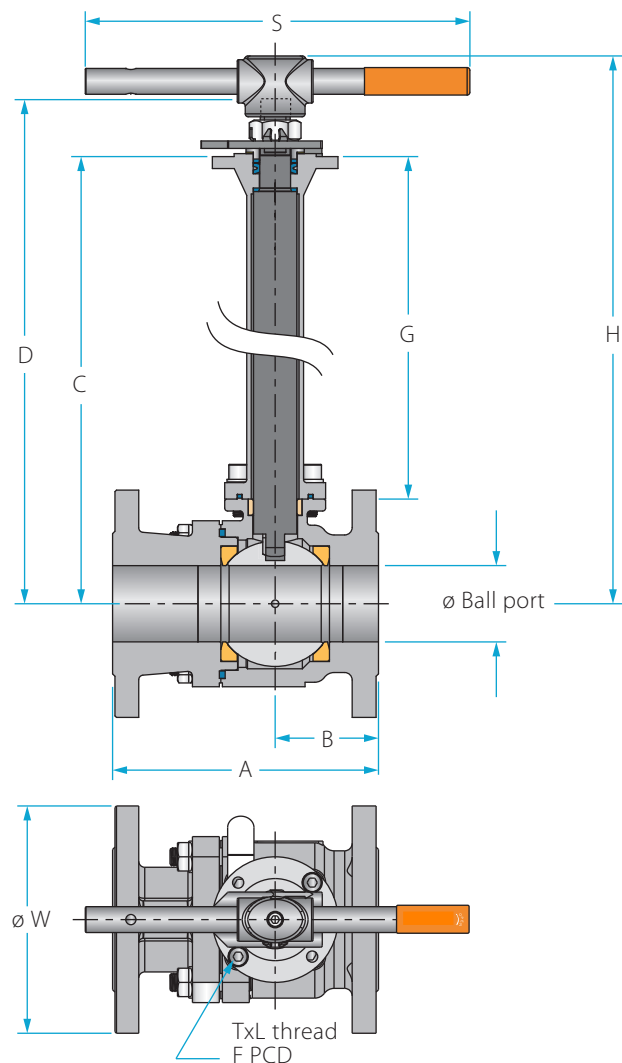
High
pressure

Multiport

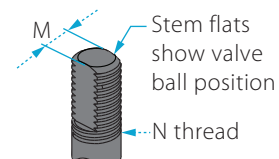
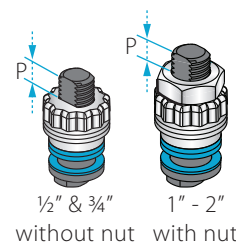
Ordering
code system

Size 1½"- 2" | DN40-DN50 | DIN PN40 | C78W/C78C⁽¹⁾ Series

Valve dimensions

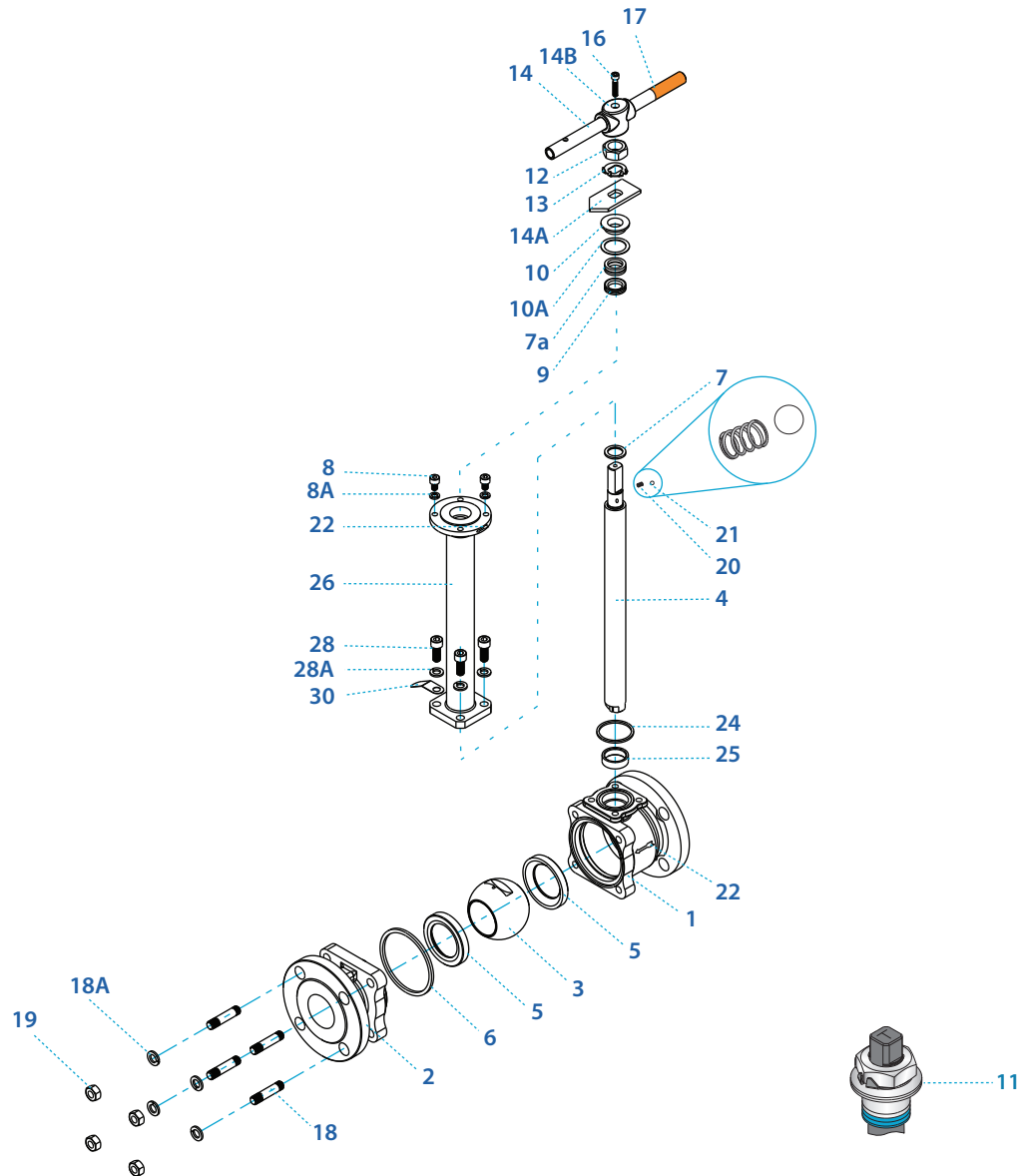


Preparation for actuation



Full Port	Unit	Ball port	A		B	Standard design				6.0" bonnet				S	W	M	N	P	F	TxL	Weight kg/lb				Kc	
						Std.		6.0"		F1	F4	F1	F4								Cv					
			F1	F4	C	D	G	H	C					D	G	H										
DN40	mm	31.80	200.00	-	65.90	352.25	381.75	304.00	406.60	200.65	230.15	152.40	255.00	237.00	150.00	8.71	9/16" UNF	8.50	(F05)	50.00	M6X8	12.8	-	12.23	-	214
1½"	inch	1.25	7.87	-	2.59	13.87	15.03	11.97	16.01	7.90	9.06	6.00	10.04	9.33	5.91	0.34		0.33		1.97		5.81	-	5.55	-	250
DN50	mm	51.00	230.00	150.00	60.70	397.50	439.10	327.50	465.10	222.40	264.00	152.40	290.00	400.00	165.00	13.90	M20X2.5	13.50	(F07)	70.00	M8X8	16.15	14.15	14.70	12.70	410
2"	inch	2.01	9.06	5.91	2.39	15.65	17.29	12.89	18.31	8.76	10.39	6.00	11.42	15.75	6.50	0.55		0.53		2.76		7.33	6.42	6.67	5.76	480

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop bolt	A582 303	2
8A	Spring washer	DIN 127 A2	2
9*	Stem seal	TFM, Graphite	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
12	Disc spring	A693 Gr. 631 17-7PH	2
13	Tab lock washer	A240 304	1

⁽¹⁾ C78C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C78C design.

Item	Description	Material specifications	Qty.
14	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	4
18A	Spring washer	DIN 127 A2	4
20	Anti-static spring	A313 302	1
21	Anti-static plunger	A479 304	1
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

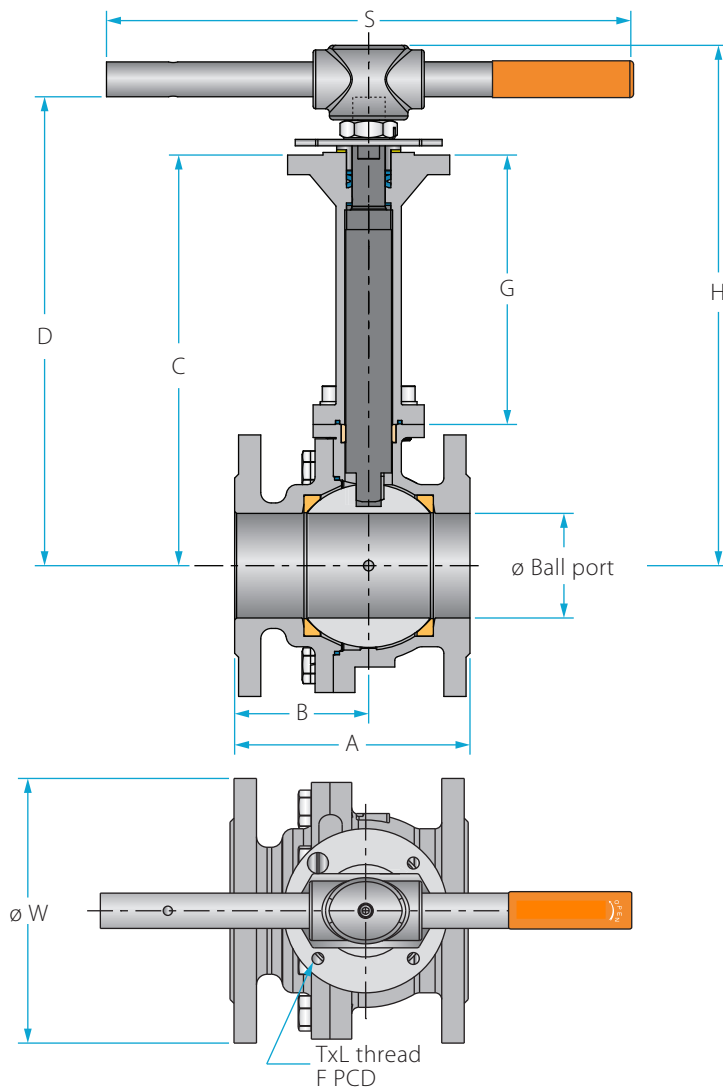
High
pressure

Multiport

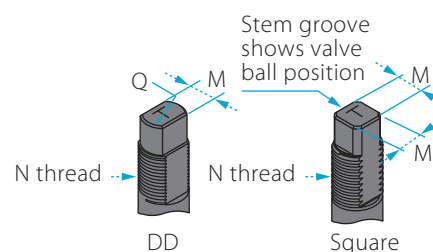
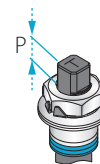
Ordering
code system

Size 3" - 6" | DN80-DN150 | DIN PN16 | C77W/C77C⁽¹⁾ Series

Valve dimensions



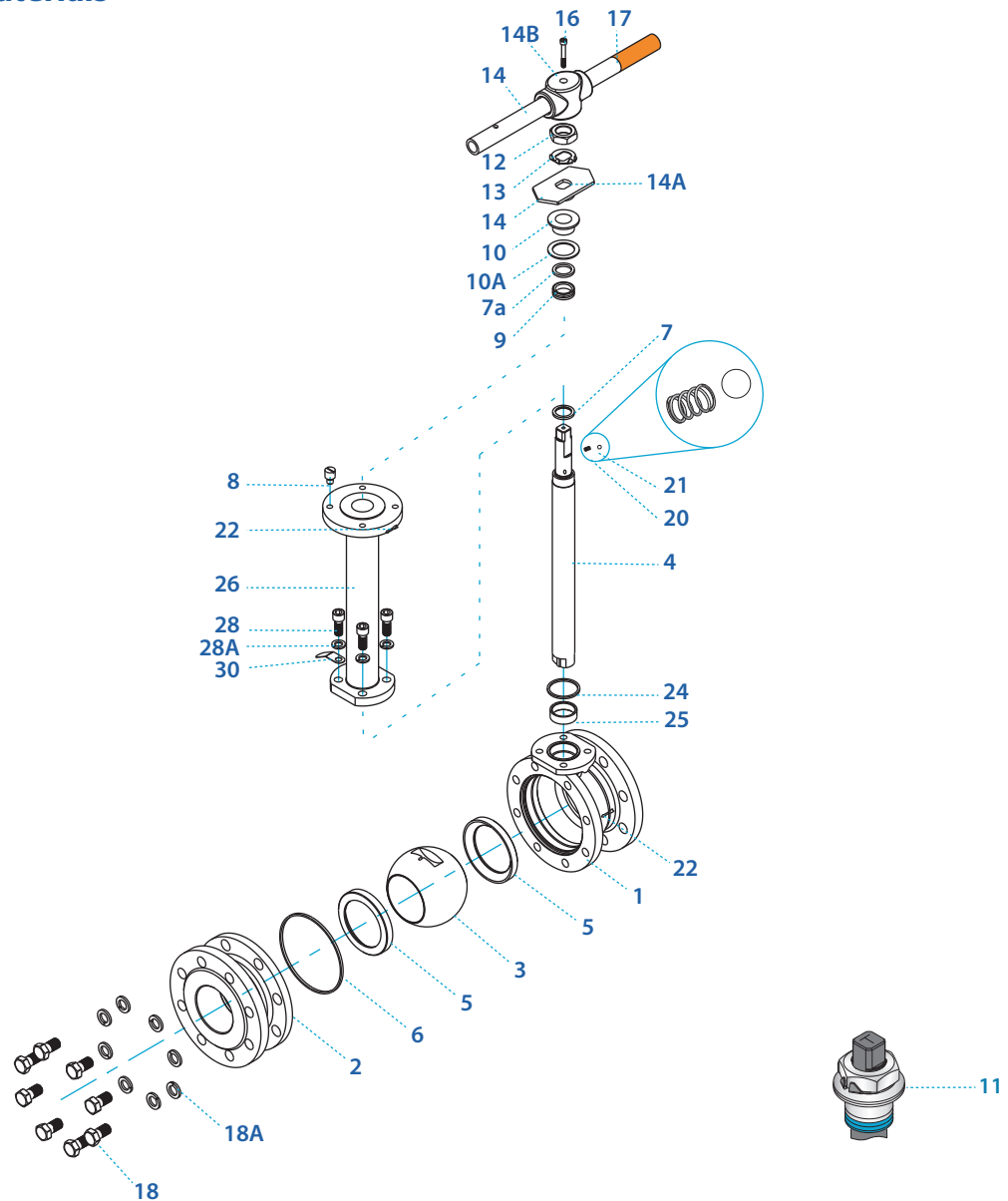
Preparation for actuation



Full Port	Unit	Ball port	A		B	Standard design				6.0" bonnet				S	W	M	M DD	N	P	F	TxL	weight kg/lb		Kv	
			F1,F5	F4		C	D	G	H	C	D	G	H									F1,F5	F4		Cv
DN80	mm	80.00	310.00	180.00	102.50	451.50	498.10	343.50	539.00	260.40	307.00	152.40	347.90	610.00	200.00	18.90	22.70	1"-14	16.70	(F10)	102.00	M10X15	31.50	30.30	1111
3"	inch	3.15	12.20	7.09	4.04	17.78	19.61	13.52	21.22	10.25	12.09	6.00	13.70	24.02	7.87	0.74	0.89	UNS-2A	0.66		4.02		69.43	66.48	1300
DN100	mm	100.00	350.00	190.00	105.50	467.50	514.10	343.50	555.00	276.40	323.00	152.40	363.90	610.00	220.00	18.90	22.70	1"-14	16.70	(F12)	125.00	M12X15	45.50	44.30	2051
4"	inch	3.94	13.78	7.48	4.15	18.41	20.24	13.52	21.85	10.88	12.72	6.00	14.33	24.02	8.66	0.74	0.89	UNS-2A	0.66		4.92		100.28	97.64	2400
DN150	mm	150.00	350.00	OPT	196.00	564.00	633.50	385.00	-	331.40	400.90	152.40	-	-	285.00	28.45	23.75	1-½"-12	26.20	(F12)	125.00	M12X15	85.00	81.00	4615
6"	inch	5.91	13.78	OPT	7.72	22.20	24.94	15.16	-	13.05	15.78	6.00	-	-	11.22	1.12	0.94	UNF-2A	1.03		4.92		187.00	178.20	5400

⁽¹⁾ Due to high valve torque, pipe handle cannot be used. A manual gear or automation means should be used to operate the valve.

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	End	A351 CF8M, A494 M35-1, Bronze RG5	1
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	1
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 Gr. 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 8M	1
13	Tab lock washer	A240 304	1
14***	Handle	C.St. Zinc plate, A240 316L	1

Item	Description	Material specifications	Qty.
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
15	Tab lock washer	A240 304	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 GrB8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN3506-1 A2-70, A193 B8	8-10
18A	Spring washer	DIN 127 A2	8-10
20	Anti-static spring	A313 302	2
21	Anti-static plunger	A479 304	2
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

⁽¹⁾ C77C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C77C design.

* Repair kit components

** Only with HermetiX™ fire safe design

*** Gear operator should be used for size 6" DN150 (handle components are not included)

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

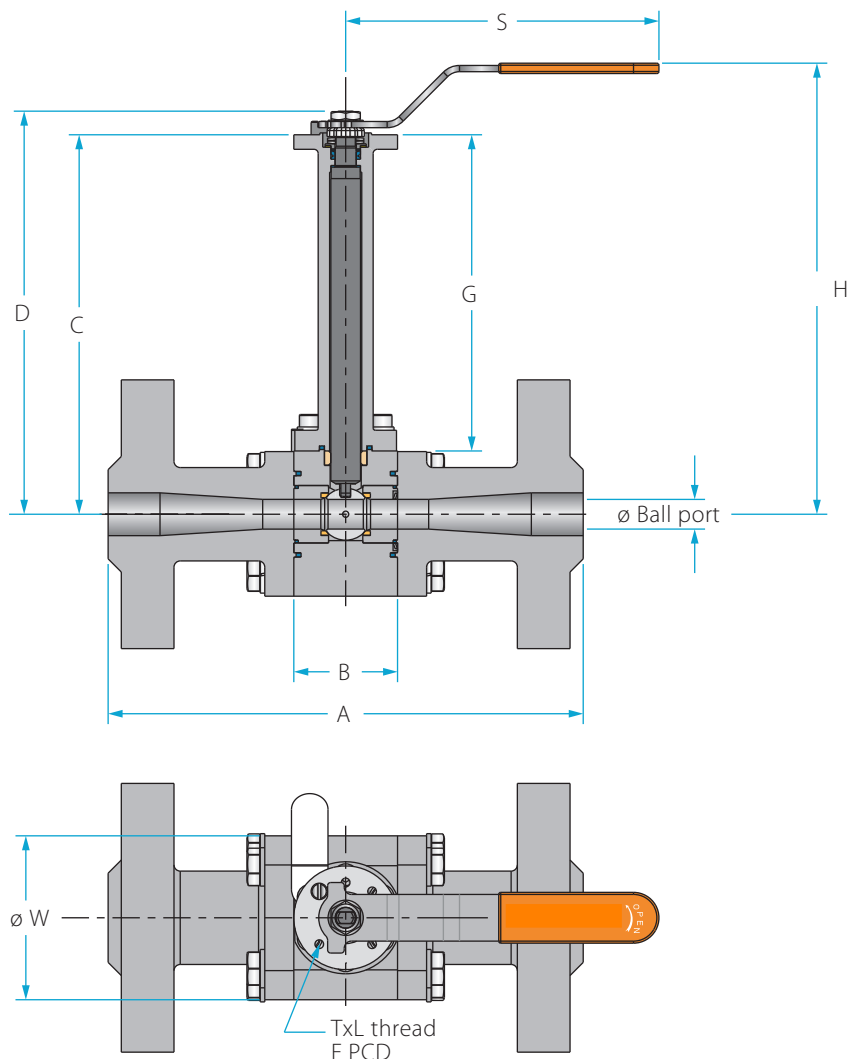
High
pressure

Multiport

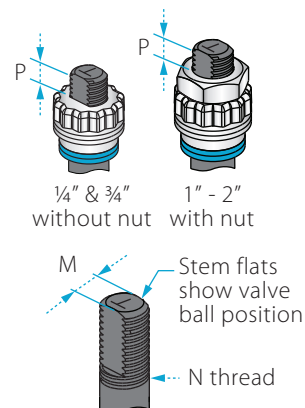
Ordering
code system

Size 1/4" - 1 1/4" | DN8-DN32 | ANSI Class 2500 | C28W/C28C⁽¹⁾ Series

Valve dimensions



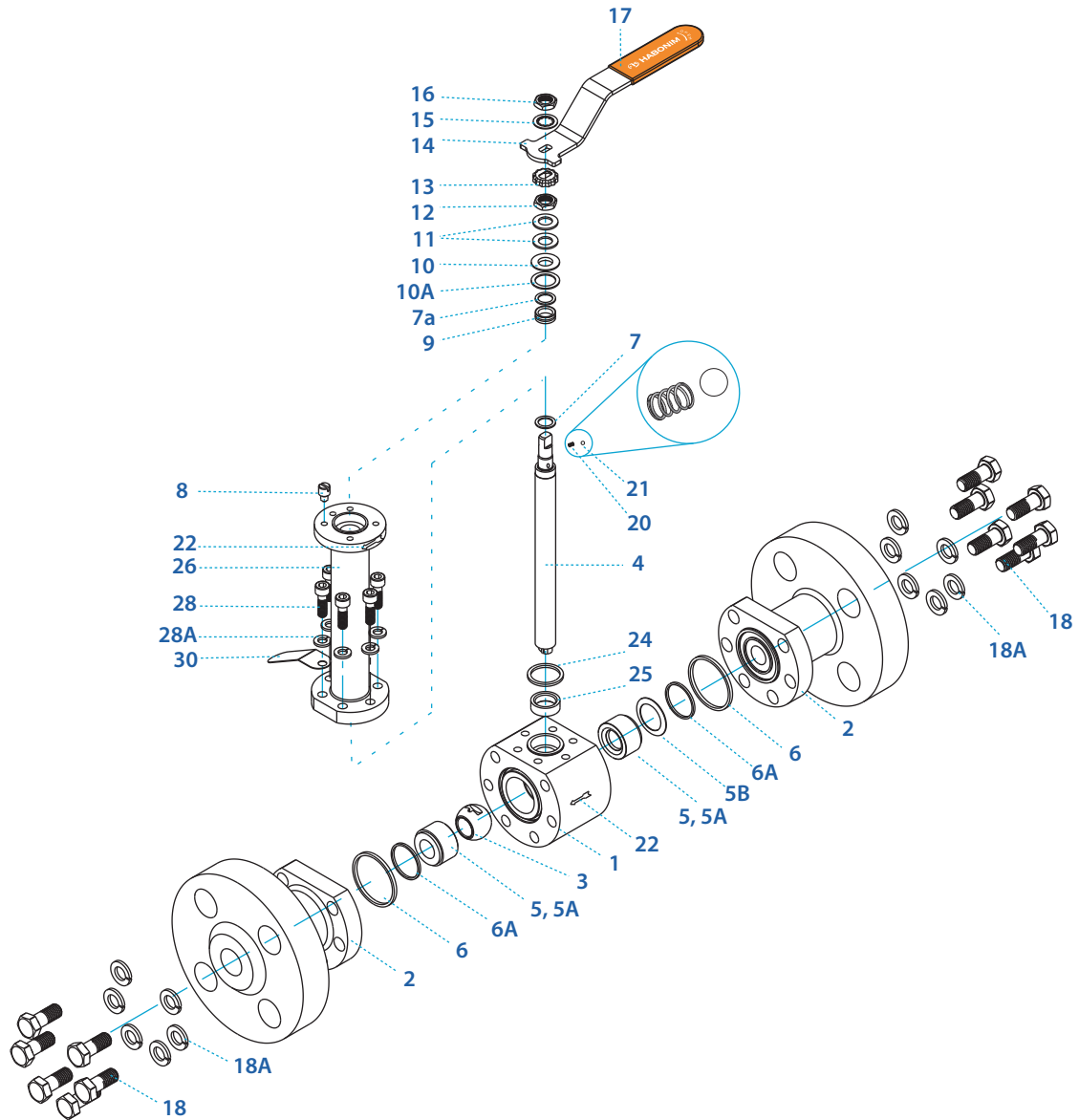
Preparation for actuation



Std. port	Full port	Unit	Ball port	A				B	Standard design				6.0" bonnet				S	W	M	N	P	F	TxL	Kv Cv	
				Welded	Threaded	#1500	#2500		C	D	G	H	C	D	G	H									
DN15	DN8, DN10	mm	11.15	173.60	107.20	216.00	264.00	50.00	333.00	341.90	304.00	380.60	181.40	190.30	152.40	229.00	150.00	78.00	5.50	3/8"	7.20	F03	36.00	M5X8	6.9
1/2"	3/8"	inch	0.44	6.83	4.22	8.50	10.39	1.97	13.11	13.46	11.97	14.98	7.14	7.49	6.00	9.02	5.91	3.07	0.22	UNF	0.28		1.42		8
DN20	DN15	mm	13.00	188.60	120.00	229.00	273.00	60.00	338.10	355.60	304.00	386.60	186.50	204.00	152.40	235.00	187.00	98.00	7.50	7/16"	7.20	F04	42.00	M5X7	10
3/4"	1/2"	inch	0.51	7.43	4.72	9.02	10.75	2.36	13.31	14.00	11.97	15.22	7.34	8.03	6.00	9.25	7.36	3.86	0.30	UNF	0.28		1.65		12
DN25	DN20	mm	20.65	208.60	144.20	254.00	308.00	68.00	347.75	377.25	304.00	401.60	196.15	225.65	152.40	250.00	237.00	114.00	8.65	9/16"	7.20	F05	50.00	M6X8	28
1"	3/4"	inch	0.81	8.21	5.68	10.00	12.13	2.68	13.69	14.85	11.97	15.81	7.72	8.88	6.00	9.84	9.33	4.49	0.34	UNF	0.28		1.97		32
DN32	DN25	mm	25.50	211.60	152.00	279.00	349.00	68.00	355.50	385.00	304.00	410.00	203.90	233.40	152.40	258.40	237.00	124.00	8.65	9/16"	8.00	F05	50.00	M6X8	49
1 1/4"	1"	inch	1.00	8.33	5.98	10.98	13.74	2.68	14.00	15.16	11.97	16.14	8.03	9.19	6.00	10.17	9.33	4.88	0.34	UNF	0.31		1.97		

* Data in the chart refers ANSI 2500 Flanges

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A479 316/316L, B574 N06022	1
2	Ends	A479 316/316L, B574 N06022	2
3	Ball	A351 CF8M, B574 N06022	1
4	Stem	A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	PCTFE (KEL-F)	2
5A*	Seat housing	A479 316/316L, B574 N06022	2
5B ⁽³⁾	Upstream seat spring	Inconel 718	1
6*	Body seal	PTFE, Graphite	2
6A*	Seat seal	PTFE, Graphite	2
7*	Stem thrust seal	PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 631 17-7PH	2

⁽¹⁾ C28C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C28C design.

⁽³⁾ Spring loaded upstream seat is Habonim default design

Item	Description	Material specifications	Qty.
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	2
13	Locking clip	A167 304	1
14	Handle	A240 430	1
15	Serrated washer	A240 410	1
16	Handle nut	EN3506-2 A4-80, A194 Gr 8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	8-16
18A	Spring washer	DIN 912 A2-70	8-16
20	Anti-static spring	A313 302	2
21	Anti-static plunger	A479 304	2
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316/316L, B574 N06022	1
28	Bolt	EN3506-1 A2-70, A193 B8	4-6
28A	Spring washer	DIN 127 A2	4-6
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

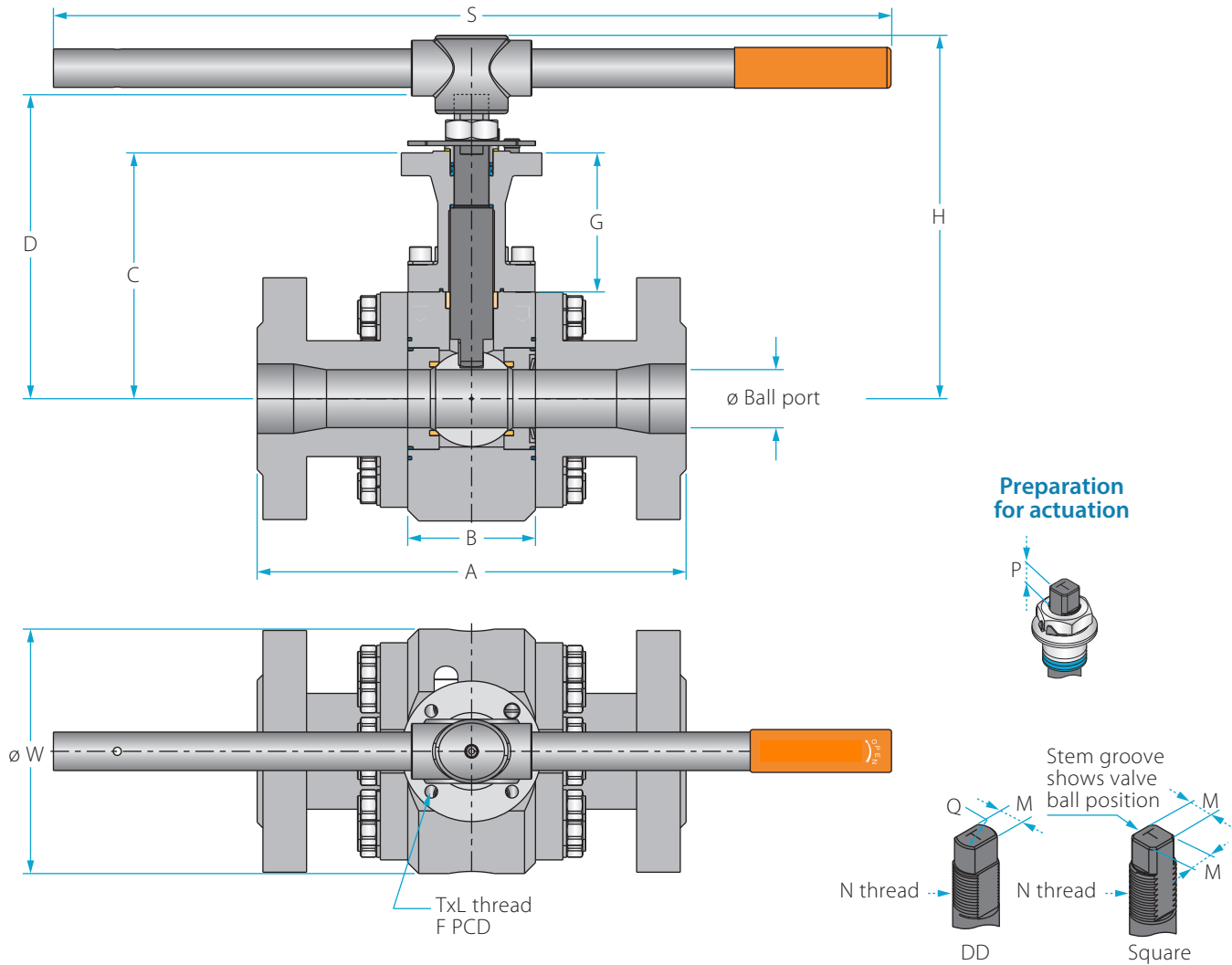
High
pressure

Multiport

Ordering
code system

Size 1½" - 8" | DN40-DN200 | ANSI Class 2500 | C28W/C28C⁽¹⁾ Series

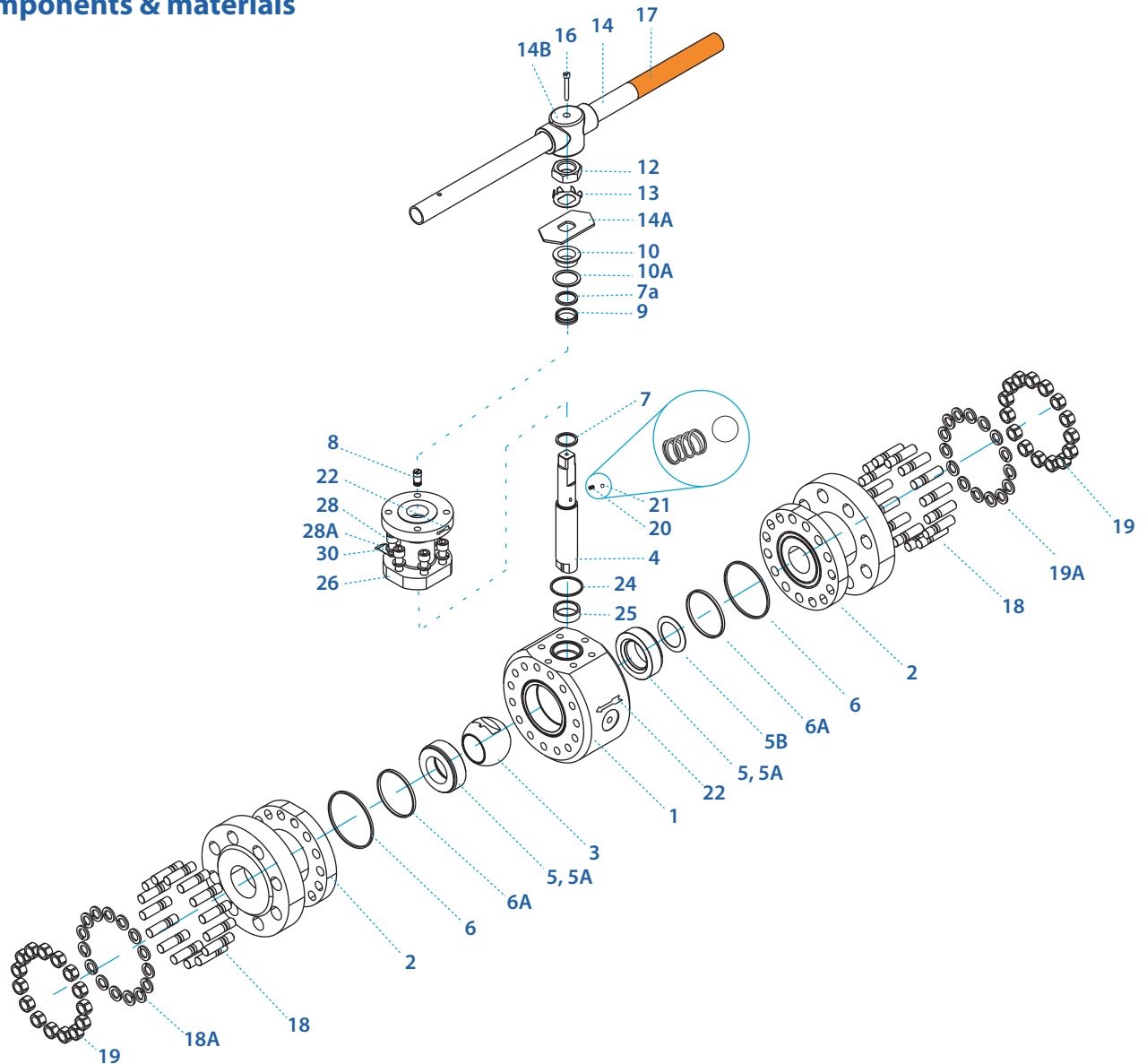
Valve dimensions



Std. port	Full port	Unit	Ball port	A				B	Standard design				6.0" bonnet				S	W	M	MDD	N	P	Q	F	TxL	Kv Cv		
				Welded	Threaded	#1500	#2500		C	D	G	H	C	D	G	H												
DN40	DN32	mm	31.8	262	179	305	384	95	357	398.6	304	425	205.4	247	152.4	273.4	400	140	13.9	13.90	M20X2.5		8	20.00	(F07)	70	M8X7	69
1½"	1¼"	inch	1.25	10.31	7.05	12.01	15.12	3.74	14.06	15.69	11.97	16.73	8.09	9.72	6.00	10.76	15.75	5.51	0.55	0.55			0.31	0.79		2.76		
DN50	DN40	mm	38.15	255.80	195.00	368.00	451.00	95.00	376.50	418.10	304.00	443.75	224.90	266.50	152.40	292.15	400.00	172.00	13.90	13.90	M20X2.5		15.75	20.00	(F07)	70.00	M8X10	102
2"	1½"	inch	1.50	10.07	7.68	14.49	17.76	3.74	14.82	16.46	11.97	17.47	8.85	10.49	6.00	11.50	15.75	6.77	0.55	0.55			0.62	0.79		2.76		
DN65	DN50	mm	51.00	313.00	205.00	419.00	508.00	105.00	392.40	438.90	304.80	479.40	240.00	286.50	152.40	327.00	610.00	199.00	18.90	15.90	1"-14 UNS-2A		16.70	22.70	(F10)	102.00	M10X15	208
2½"	2"	inch	2.01	12.32	8.07	16.50	20.00	4.13	15.45	17.28	12.00	18.87	9.45	11.28	6.00	12.87	24.02	7.83	0.74	0.63			0.66	0.89		4.02		
DN80	DN65	mm	63.75	343.20	-	470.00	578.00	140.00	421.00	490.10	304.00	-	269.40	338.50	152.40	-	-	268.00	28.45	23.75	1-½" UNF-2A		26.20	35.20	(F12)	125.00	M16X25	300
3"	2½"	inch	2.51	13.51	-	18.50	22.76	5.51	16.57	19.30	11.97	-	10.61	13.33	6.00	-	-	10.55	1.12	0.94			1.03	1.39		4.92		
DN100	DN80	mm	82.80	419.20	-	546.00	673.00	140.00	438.00	550.65	304.00	-	286.40	399.05	152.40	-	-	298.00	28.45	23.75	1-½" UNF-2A		26.20	35.20	(F12)	125.00	M16X25	615
4"	3"	inch	3.26	16.50	-	21.50	26.50	5.51	17.24	21.68	11.97	-	11.28	15.71	6.00	-	-	11.73	1.12	0.94			1.03	1.39		4.92		
DN150	DN100	mm	100.00	510.00	-	705.00	914.00	175.00	451.00	563.65	304.00	-	299.40	412.05	152.40	-	-	339.00	35.90	35.90	2" UNF-2A		40.00	46.50	(F16)	165.00	M20X25	872
6"	4"	inch	3.94	20.08	-	27.76	35.98	6.89	17.76	22.19	11.97	-	11.79	16.22	6.00	-	-	13.35	1.41	1.41			1.57	1.83		6.50		
DN200	DN150	mm	144.40	680.00	-	832.00	1022.00	210.00	496.00	637.00	304.00	-	344.40	485.40	152.40	-	-	423.00	45.90	45.90	2-¾" UN-2A		40.00	59.00	(F16)	165.00	M20X25	1552
8"	6"	inch	5.69	26.77	-	32.76	40.24	8.27	19.53	25.08	11.97	-	13.56	19.11	6.00	-	-	16.65	1.81	1.81			1.57	2.32		6.50		

⁽¹⁾ Due to high valve torque, pipe handle cannot be used. A manual gear or automation means should be used to operate the valve.

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A479 316/316L, B574 N06022	1
2	Ends	A479 316/316L, B574 N06022	2
3	Ball	A351 CF8M, B574 N06022	1
4	Stem	A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	PCTFE (KEL-F)	2
5A*	Seat housing	A479 316/316L, B574 N06022	2
5B ⁽³⁾	Upstream seat spring	Inconel 718	1
6*	Body seal	PTFE, Graphite	2
6A*	Seat seal	PTFE, Graphite	2
7*	Stem thrust seal	PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1
13	Tab lock washer	A240 304	1
14***	Handle	C.St. Zinc plate, A240 316L	1

Item	Description	Material specifications	Qty.
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	8-16
18A	Spring washer	DIN 127 A2	8-16
19	Body bolt	EN3506-1 A2-70, A193 B8	8-16
19A	Spring washer	DIN 912 A2-70	15
20	Anti-static spring	A313 302	2
21	Anti-static plunger	A479 304	2
22	Arrow flow	A167 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316/316L, B574 N06022	1
28	Bolt	EN3506-1 A2-70, A193 B8	6-8
28A	Spring washer	DIN 127 A2	6-8
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe design

*** Cap operators should be used for size 2" DN80 (handle components are not included)

⁽¹⁾ C28C series is Habonim's standard valve design without the HermetiX™ stem

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

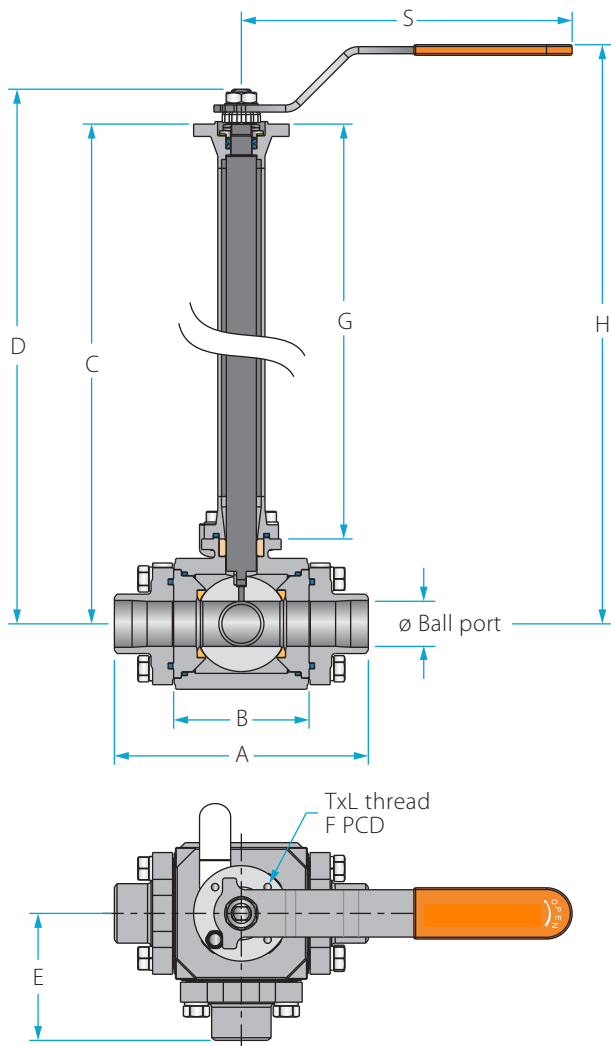
High
pressure

Multiport

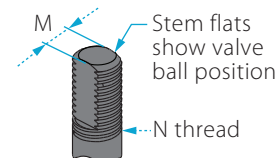
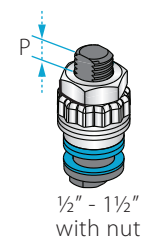
Ordering
code system

Size 1/2" - 1 1/2" | DN15-DN40 | ANSI Class 600 | C61W/C61C Series

Valve dimensions

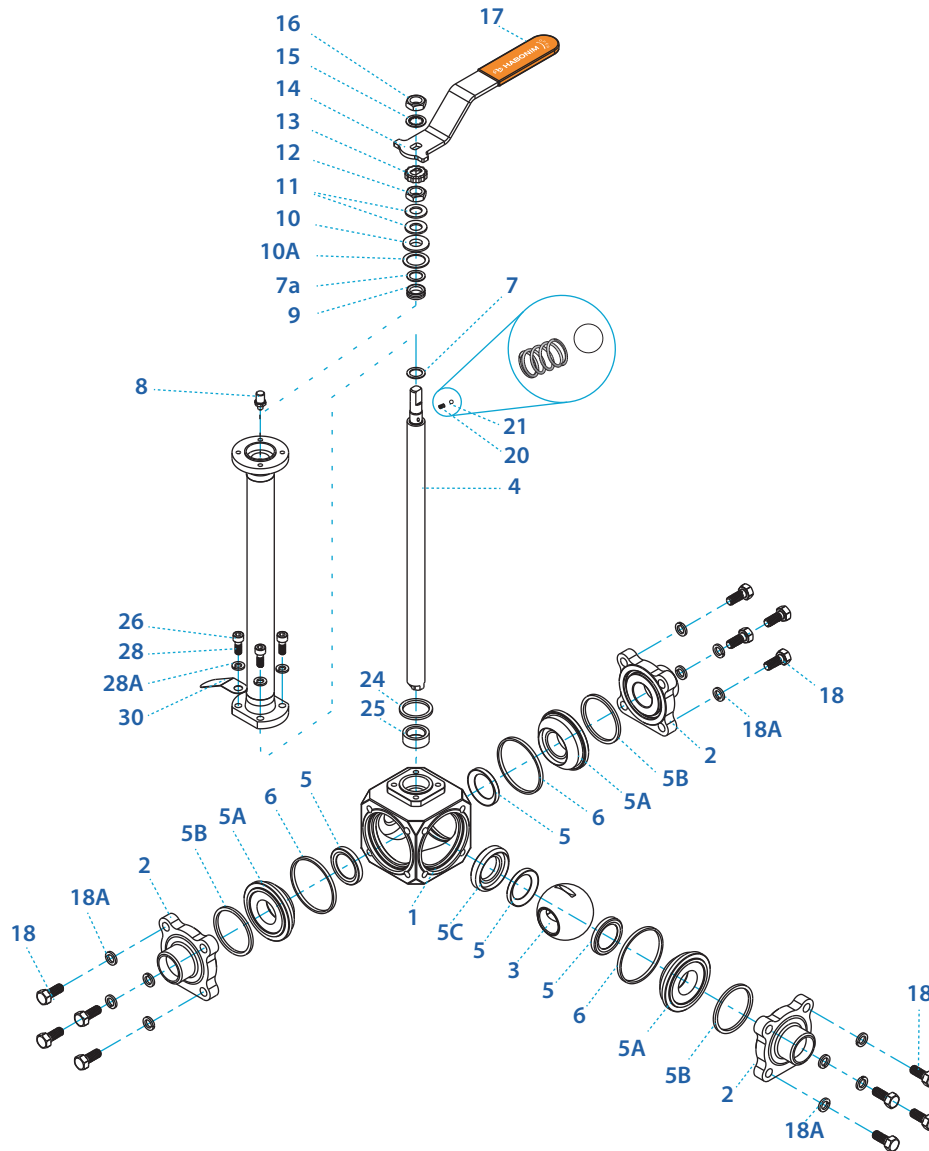


Preparation for actuation



Std. port	Full port	Unit	Ball port	A	B	Standard design					6.0" bonnet				E	S	M	N	P	F	TxL	Weight kg/lb	
						C	D	E	G	H	C	D	G	H								Std.	6.0 Bonnet
DN20	DN15	mm	14.30	108.00	62.00	341.00	358.50	304.00	384.90	189.40	206.90	152.40	233.30	224.52	54.30	187.00	7.50	7/16"	7.20 (F04)	42.00	M5X7	3.65	3.33
3/4"	1/2"	inch	0.56	4.25	2.44	13.43	14.11	11.97	15.15	7.46	8.15	6.00	9.19	8.84	2.14	7.36	0.30	UNF	0.28	1.65	M5X7	8.04	7.34
DN25	DN20	mm	20.70	134.60	71.60	347.00	364.50	304.00	389.80	195.40	212.90	152.40	238.20	239.43	66.80	187.00	7.50	7/16"	7.20 (F04)	42.00	M5X7	5.25	4.90
1"	3/4"	inch	0.81	5.30	2.82	13.66	14.35	11.97	15.35	7.69	8.38	6.00	9.38	9.43	2.63	7.36	0.30	UNF	0.28	1.65	M5X7	11.57	10.80
DN32	DN25	mm	25.50	143.00	76.00	352.00	369.50	304.00	396.90	200.40	217.90	152.40	245.30	244.43	71.80	187.00	7.50	7/16"	8.50 (F04)	42.00	M5X7	7.42	6.85
1 1/4"	1"	inch	1.00	5.63	2.99	13.86	14.55	11.97	15.63	7.89	8.58	6.00	9.66	9.62	2.83	7.36	0.30	UNF	0.33	1.65	M5X7	16.35	15.10
DN40	DN32	mm	31.65	158.00	91.00	353.40	382.90	304.00	408.10	201.80	231.30	152.40	256.50	257.13	79.00	237.00	8.71	9/16"	8.50 (F05)	50.00	M6X8	9.12	8.55
1 1/2"	1 1/4"	inch	1.25	6.22	3.58	13.91	15.07	11.97	16.07	7.94	9.11	6.00	10.10	10.12	3.11	9.33	0.34	UNF	0.33	1.97	M6X8	20.10	18.84

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	Ends	A351 CF8M, A494 M35-1, Bronze RG5	3
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE,TFM, PCTFE (KEL-F)	4
5A	Seat housing	A479 316/316L, B574 N06022, Bronze RG5	3
5B*	Seat seal	PTFE, Graphite	3
5C	Seat support	A479 316/316L, B574 N06022, Bronze RG5	1
6*	Body seal	PTFE, Graphite	3
7*	Stem thrust seal	TFM ⁽¹⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1

⁽¹⁾ C61C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C61C design.

Item	Description	Material specifications	Qty.
11	Disc spring	A693 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1
13	Locking clip	A167 304	1
14	Handle	A240 430	1
15	Serrated washer	A240 410	1
16	Handle nut	EN3506-2 A4-80, A194 Gr 8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	12
18A	Spring washer	DIN 127 A2	12
20	Anti static spring	A313 302	1
21	Anti static plunger	A479 304	1
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe stem seal

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

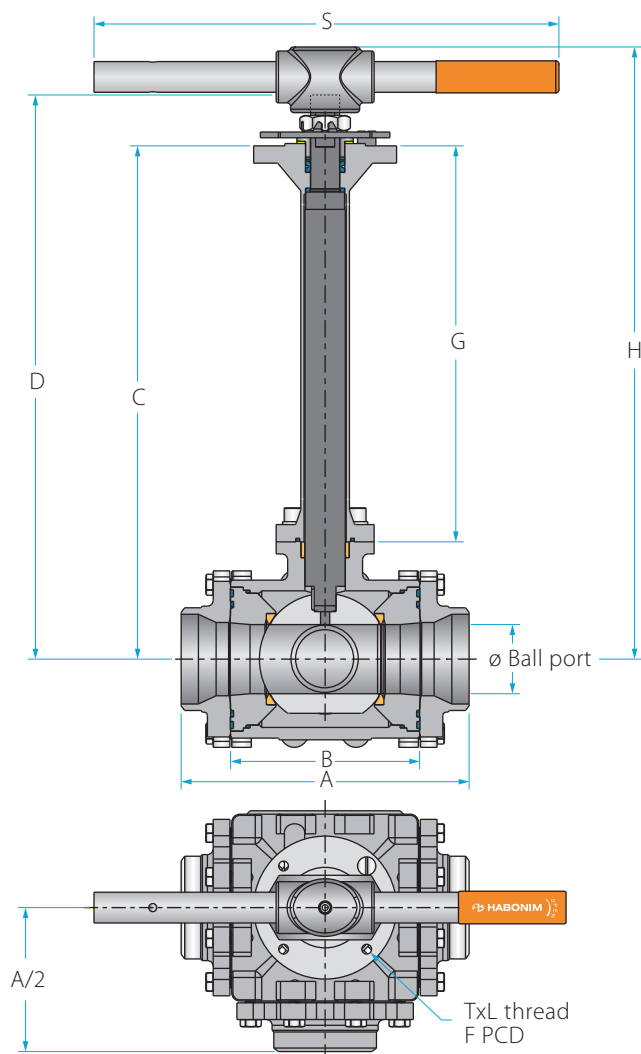
High
pressure

Multiport

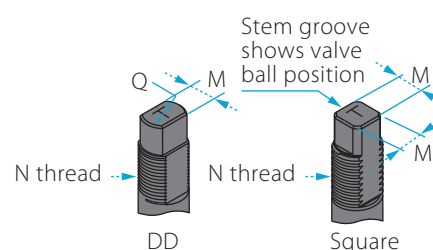
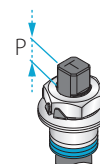
Ordering
code system

Size 2" - 4" | DN50-DN100 | ANSI Class 300 | C61W/C61C Series

Valve dimensions



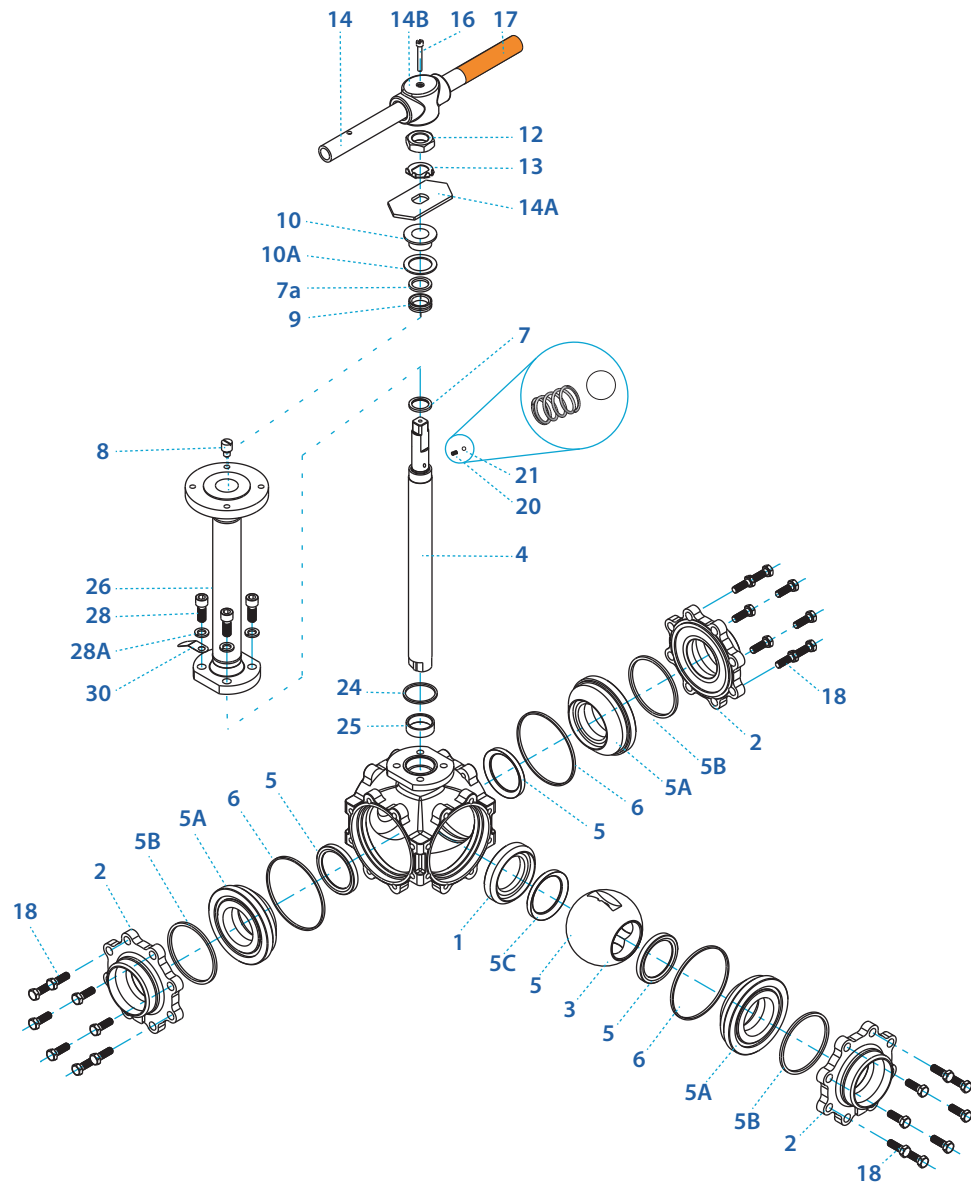
Preparation for actuation



Std. port	Full port	Unit	Ball port	A	B	Standard design					6.0" bonnet				S	M	MDD	N	P	Q	F	TxL	Weight kg/lb		
						C	D	E	G	H	C	D	G	H									Std.	6.0 Bonnet	
DN50	DN40	mm	38.10	178.80	107.20	397.10	438.70	327.50	465.90	222.00	263.60	152.40	290.80	224.52	400.00	13.90	13.90	M20X2.5	14.36	20.00	(F07)	70.00	M8X8	24.70	23.25
2"	1½"	inch	1.50	7.04	4.22	15.63	17.27	12.89	18.34	8.74	10.38	6.00	11.45	8.84	15.75	0.55	0.55		0.57	0.79		2.76		54.44	51.24
DN65	DN50	mm	48.00	210.00	125.00	406.60	453.10	327.50	480.30	231.50	278.00	152.40	305.20	239.43	400.00	18.90	15.90	1"-14 UNS-2A	16.70	22.70	(F10)	102.00	M10X15	24.70	23.25
2½"	2"	inch	1.89	8.27	4.92	16.01	17.84	12.89	18.91	9.11	10.94	6.00	12.02	9.43	15.75	0.74	0.63		0.66	0.89		4.02		54.44	51.24
DN80	DN65	mm	60.00	249.80	164.00	486.75	555.85	385.00	-	254.15	323.25	152.40	-	244.43	-	28.45	23.75	1-½"-12 UNF-2A	26.20	35.20	(F12)	125.00	M12X15	42.40	41.20
3"	2½"	inch	2.36	9.83	6.46	19.16	21.88	15.16	-	10.01	12.73	6.00	-	9.62	-	1.12	0.94		1.03	1.39		4.92		93.45	90.80
DN100	DN80	mm	76.00	309.80	205.00	519.50	588.60	385.00	-	286.90	356.00	152.40	-	257.13	-	28.45	23.75	1-½"-12 UNF-2A	26.20	35.20	(F12)	125.00	M12X15	66.40	65.20
4"	3"	inch	2.99	12.20	8.07	20.45	23.17	15.16	-	11.30	14.02	6.00	-	10.12	-	1.12	0.94		1.03	1.39		4.92		146.35	143.7

⁽¹⁾ Due to high valve torque, pipe handle cannot be used. A manual gear or automation means should be used to operate the valve.

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	Ends	A351 CF8M, A494 M35-1, Bronze RG5	3
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE, TFM, PCTFE (KEL-F)	4
5A	Seat housing	A479 316/316L, B574 N06022, Bronze RG5	3
5B*	Seat seal	PTFE, Graphite	3
5C	Seat support	A479 316/316L, B574 N06022, Bronze RG5	1
6*	Body seal	PTFE, Graphite	3
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1

⁽¹⁾ C61C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the C61C design.

Item	Description	Material specifications	Qty.
13	Tab lock washer	A240 304	1
14***	Handle	C.St. Zinc plate, A240 316L	1
14A	Stop plate	A240 430	1
14B	Wrench head	A351 CF8M	1
16	Wrench bolt	EN3506-1 A2-70/A4-80, A193 Gr B8/B8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	12-24
18A	Spring washer	DIN 127 A2	12-24
20	Anti static spring	A313 302	2
21	Anti static plunger	A479 304	2
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe stem seal

*** Gear operator should be used for size 3" DN80 and above (handle components are not included)

Floating Ball Valves

Cryogenic valves

General

3 Piece

Flanged

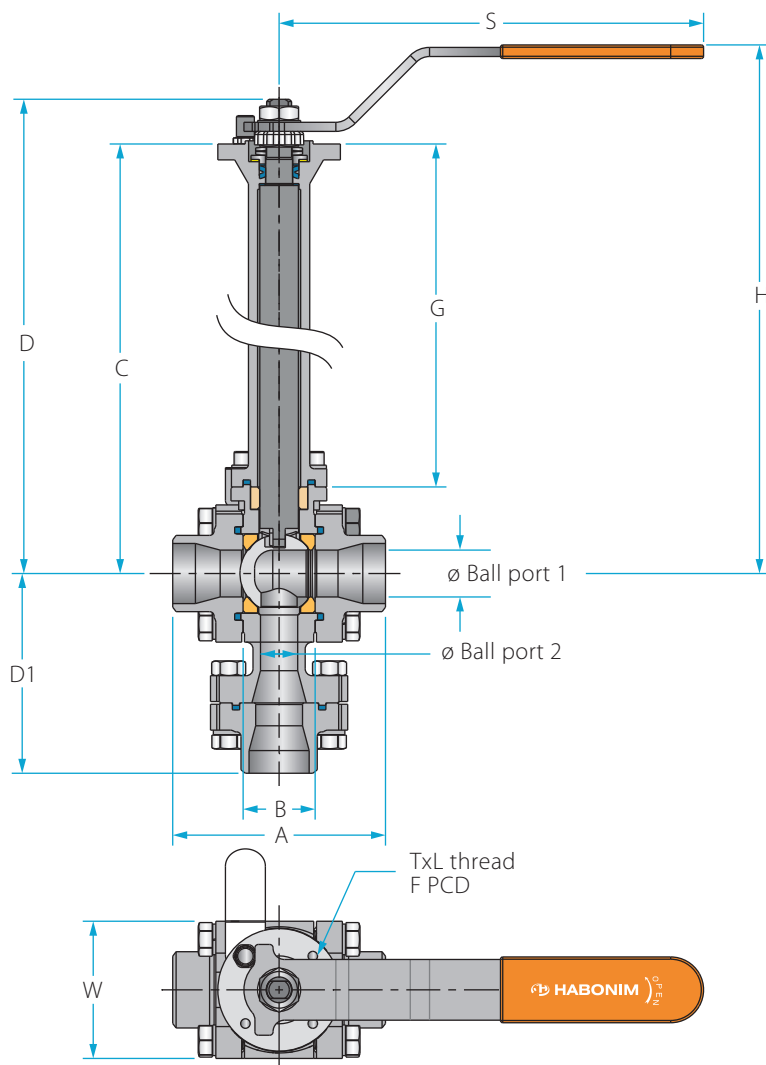
High
pressure

Multiport

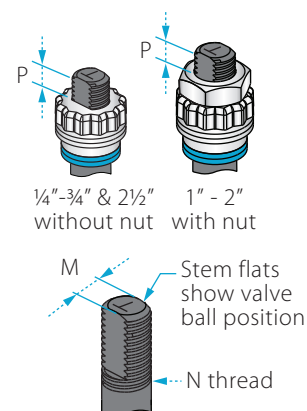
Ordering
code system

Size 1/2" - 2" | DN15-DN50 | ANSI Class 600 | DC47W Series

Valve dimensions

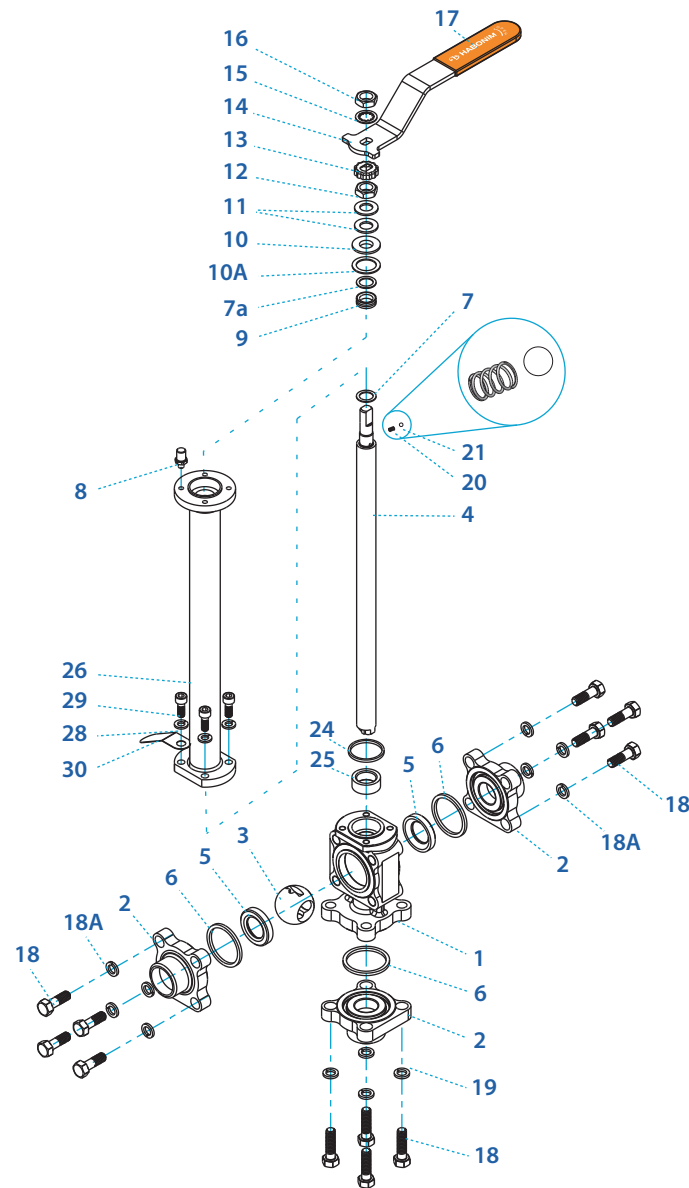


Preparation for actuation



Std. port	Full Port	Unit	Ball port1	Ball port2	A	B	Standard design					6.0" bonnet					S	W	M	N	P	F	TxL	Weight kg/lb	
							C	D	K	G	H	C	D	K	G	H								Std.	6.0 Bonnet
DN15	DN10	mm	10.50	9.50	65.80	20.60	333.00	341.90	304.00	367.30	181.40	190.30	152.40	215.70	152.80	214.30	150.00	45.80	5.50	3/8" UNF	7.20 (F03)	36.00	MSX7	1.79	1.47
1/2"	3/8"	inch	0.41	0.37	2.59	0.81	13.11	13.46	11.97	14.46	7.14	7.49	6.00	8.49	6.02	8.44	5.91	1.80	0.22	3/8" UNF	0.28	1.42	MSX7	3.93	3.23
DN20	DN15	mm	14.30	11.90	70.55	24.55	335.40	344.30	304.00	369.60	183.80	192.70	152.40	218.00	152.80	216.70	150.00	52.10	5.50	3/8" UNF	7.20 (F03)	36.00	MSX7	1.52	1.20
3/4"	1/2"	inch	0.56	0.47	2.78	0.97	13.20	13.56	11.97	14.55	7.24	7.59	6.00	8.58	6.02	8.53	5.91	2.05	0.22	3/8" UNF	0.28	1.42	MSX7	3.35	2.64
DN25	DN20	mm	20.65	15.90	93.75	31.75	342.15	359.65	304.00	387.00	190.55	208.05	152.40	235.40	152.80	232.20	187.00	60.45	7.54	7/16" UNF	7.20 (F04)	42.00	MSX7	3.19	2.84
1"	3/4"	inch	0.81	0.63	3.69	1.25	13.47	14.16	11.97	15.24	7.50	8.19	6.00	9.27	6.02	9.14	7.32	2.38	0.30	7/16" UNF	0.28	1.65	MSX7	7.02	6.25
DN40	DN25	mm	26.60	31.80	115.35	48.35	347.55	377.05	304.00	404.00	195.95	225.45	152.40	252.40	152.80	249.80	237.00	79.15	8.71	9/16" UNF	8.00 (F05)	50.00	M6X8	5.22	4.65
1 1/2"	1"	inch	1.05	1.25	4.54	1.90	13.68	14.84	11.97	15.91	7.71	8.88	6.00	9.94	6.02	9.83	9.29	3.12	0.34	9/16" UNF	0.31	1.97	M6X8	11.51	10.25
DN50	DN40	mm	38.15	35.00	127.90	56.30	352.25	381.75	304.00	409.00	200.65	230.15	152.40	257.40	152.80	254.60	237.00	90.70	8.71	9/16" UNF	8.50 (F05)	50.00	M6X8	6.68	6.11
2"	1 1/2"	inch	1.50	1.38	5.04	2.22	13.87	15.03	11.97	16.10	7.90	9.06	6.00	10.13	6.02	11.96	9.29	3.57	0.34	9/16" UNF	0.33	1.97	M6X8	14.72	13.47

Components & materials



Item	Description	Material specifications	Qty.
1	Body	A351 CF8M, A494 M35-1, Bronze RG5	1
2	Ends	A351 CF8M, A494 M35-1, Bronze RG5	3
3	Ball	A351 CF8M, B574 N06022, Bronze RG5	1
4	Stem	A479 316/316L, A564 Gr.630 H1150D 17-4PH, B574 N06022	1
5*	Seat	CF PTFE,TFM, PCTFE (KEL-F)	2
6*	Body seal	PTFE, Graphite	3
7*	Stem thrust seal	TFM ⁽²⁾ , PCTFE (KEL-F)	1
7a*	Anti-abrasion ring	TFM ⁽²⁾ , PCTFE (KEL-F)	1
8	Stop pin	A582 303	1
9*	Stem seal	TFM, Graphite ⁽²⁾	1
10	Follower	B783 316L	1
10A**	Slide bearing	TF316	1
11	Disc spring	A693 631 17-7PH	2
12	Stem nut	EN3506-2 A4-80, A194 Gr 8M	1
13	Locking clip	A167 304	1
14	Handle	A240 430	1

⁽¹⁾ DC47C series is Habonim's standard valve design without the HermetiX™ stem seal construction.

⁽²⁾ This material can only be used as part of the DC47C design.

Item	Description	Material specifications	Qty.
15	Serrated washer	A240 410	1
16	Handle nut	EN3506-2 A4-80, A194 Gr 8M	1
17	Sleeve	PVC	1
18	Body bolt	EN 3506-1 A2-70, A193 Gr B8	12
18A	Spring washer	DIN 127 A2	12
20	Anti static spring	A313 302	1
21	Anti static plunger	A479 304	1
23	Tag (not shown)	A167 304	1
24	Bonnet seal	PTFE, Graphite	1
25	Stem bearing	PTFE	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	12
28A	Spring washer	DIN 127 A2	12
30	Dog tag	A167 304	1
26	Bonnet	A479 316L, A351 CF8M	1
28	Bolt	EN3506-1 A2-70, A193 B8	4
28A	Spring washer	DIN 127 A2	4
30	Dog tag	A167 304	1

* Repair kit components

** Only with HermetiX™ fire safe stem seal